Fluid transport

1. Learning objectives
2. Pump classification
   - Pump curves examples
3. System curve
4. Pump selection
5. Cavitation
6. NPSH
7. Affinity laws
8. Design practice-Summary
9. Practice problem

Learning objectives

You will learn about the

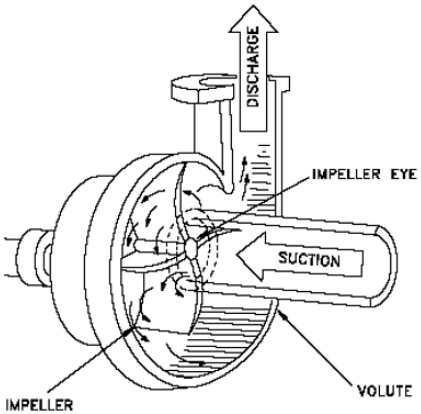
- Pump classification
- Pump performance curve
- Best efficiency point
- System curve
- Pump selection

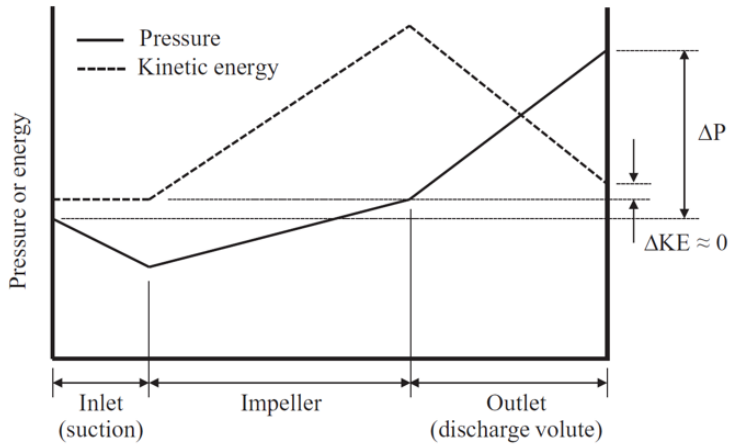
Pump types

- Positive displacement
  - Reciprocating action
    - Energy transfer to the fluid by pistons, plungers, diaphragms, or bellows
  - Rotary action
    - Energy transfer to the fluid by vanes, screws, or lobes
- Dynamic
  - Centrifugal
    - Energy transferred to the fluid by a rotating impeller in the radial direction
  - Propeller action
    - Energy transferred to the fluid by a rotating impeller in the axial direction
  - Mixed action
    - Energy transferred to the fluid by a combination of radial and axial motion through an impeller

Operation- Dynamic pump

- Fluid enters the pump through the suction port
  - There is a small drop in pressure due to friction and the kinetic energy stays constant because the suction line diameter is constant
- Fluid moves through the impeller
  - The fluid experiences a significant increase in velocity and the pressure begins to rise
- The largest pressure increase in the pump occurs in the discharge passage of the pump. The discharge passage is called a discharge volute.
  - The discharge volute is a liquid diffuser. The cross-sectional flow area of the volute increases in the direction of fluid flow.
- This pressure increase is offset by decrease in kinetic energy to satisfy the conservation of energy equation.
  - When the fluid finally leaves the pump, its outlet velocity is very nearly the same as its velocity at the suction port, but the pressure has increased significantly.





Centrifugal pump performance

When conducting centrifugal pump performance experiment

- Measure quantities
  - Shaft torque
  - Impeller rotational speed
  - Inlet pressure
  - Outlet pressure
  - Volumetric flow rate
  - Suction and discharge pipe specifications
- Calculate quantities
  - Pump head delivered
  - Pump power input
  - Power input to impeller
  - Pump efficiency

Centrifugal pump performance

Equations to calculate pump performance

$$H_p = H_d - H_s \tag{1}$$

$$H_p = \left(\frac{P_d}{\gamma} + \frac{V_d^2}{2g} + z_d\right) - \left(\frac{P_s}{\gamma} + \frac{V_s^2}{2g} + z_s\right) \tag{2}$$

$$\dot{W} = \dot{m} g H_p \tag{3}$$

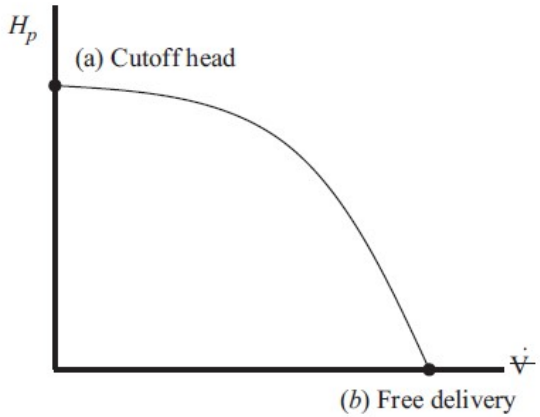
$$\eta_p = \frac{\dot{W}}{\dot{W}_{bhp}} \tag{4}$$

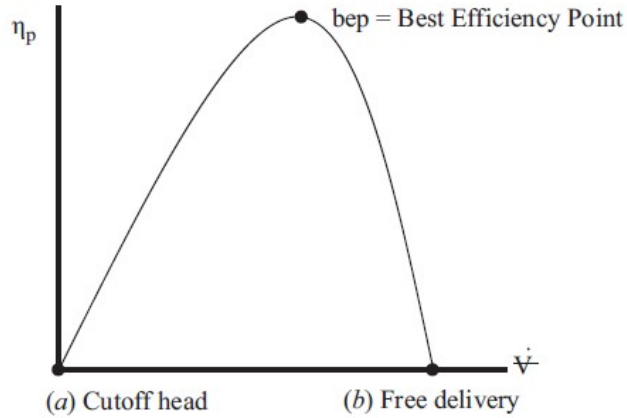
$$\dot{W}_{bhp} = T\omega \tag{5}$$

Centrifugal pump performance

Two interesting extremes can be observed on a pump performance curve. These are indicated on the plot on the left side of Figure below as the cutoff head, point (a), and the free delivery condition, point (b).
The cutoff head represents the condition where the pump is not able to provide any flow to the system. A good example of this is a pump being used to move water from a stream to an irrigation ditch. If the elevation of the ditch is higher than the cutoff head, then the pump cannot provide the flow. In fact, there will be no flow at all. For this condition, the efficiency of the pump is zero,

$$\eta p, cutoff = 0 = \frac{0 g H_p}{\dot{W}_{bhp}} \tag{6}$$

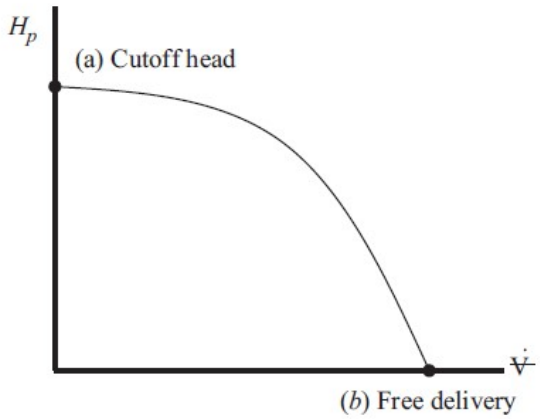


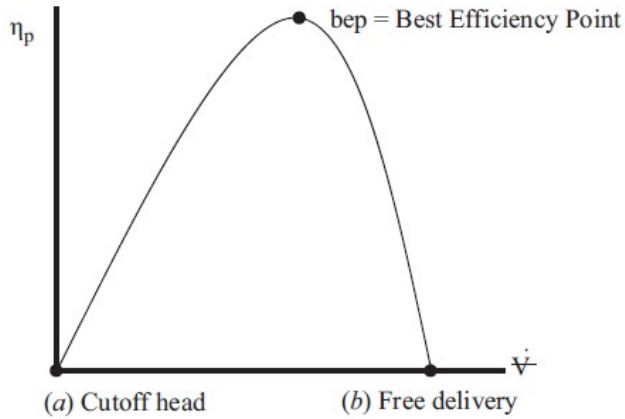


Centrifugal pump performance

The other extreme point, free delivery, occurs if the pump curves were extrapolated to the zero-head axis. In this case, the pump is providing no power input to the fluid and the fluid is passing through the pump as if it were a pipe. In this case, the efficiency is also zero,

$$\eta p, free = 0 = \frac{\dot{m} g 0}{\dot{W}_{bhp}} \tag{7}$$

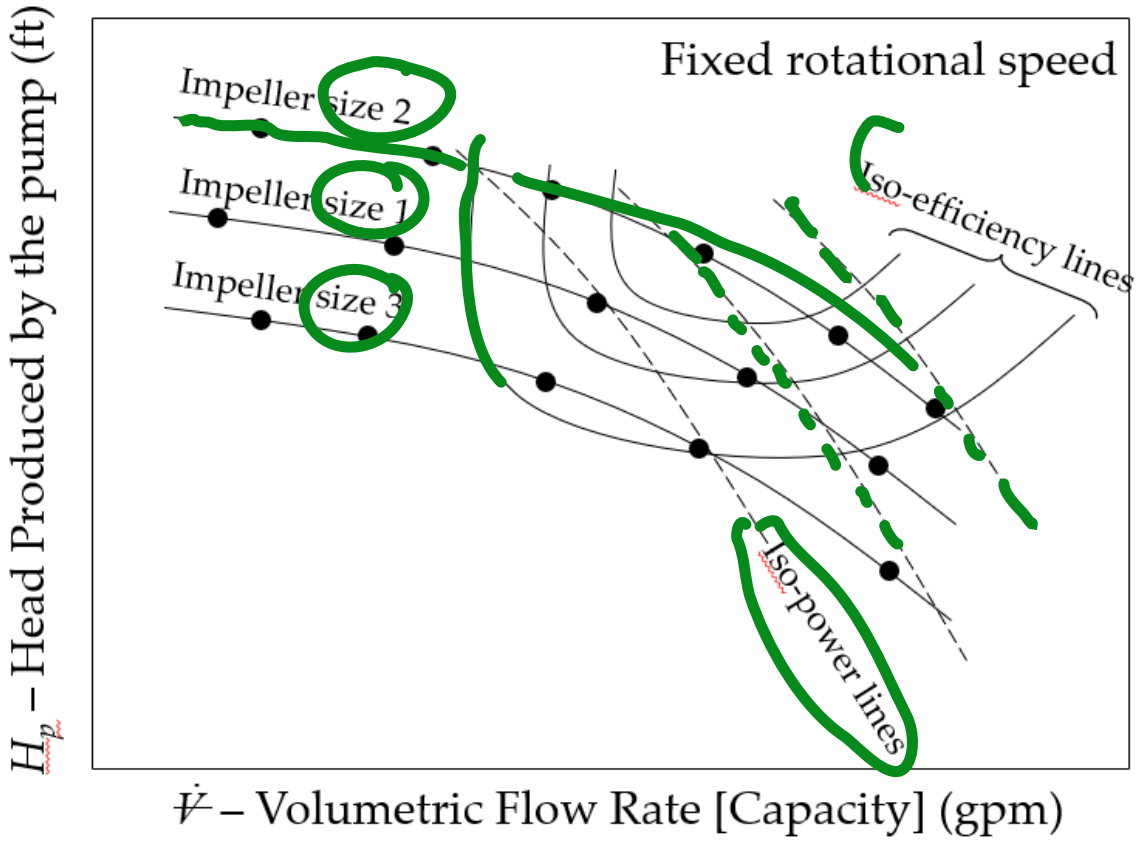




Reading and understanding pump performance

From the figure we can get

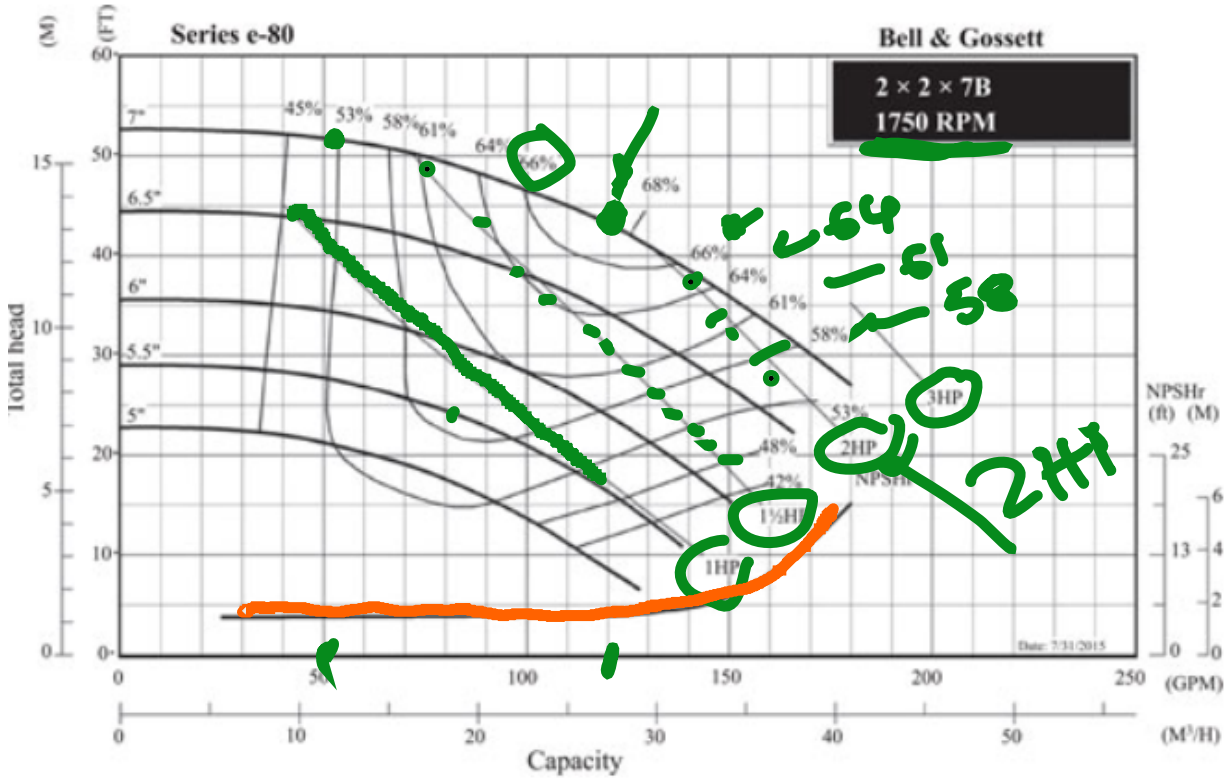
- Pump efficiency
- Pump impeller size
- Total head
- HP
- NPSH Req



Reading and understanding pump performance

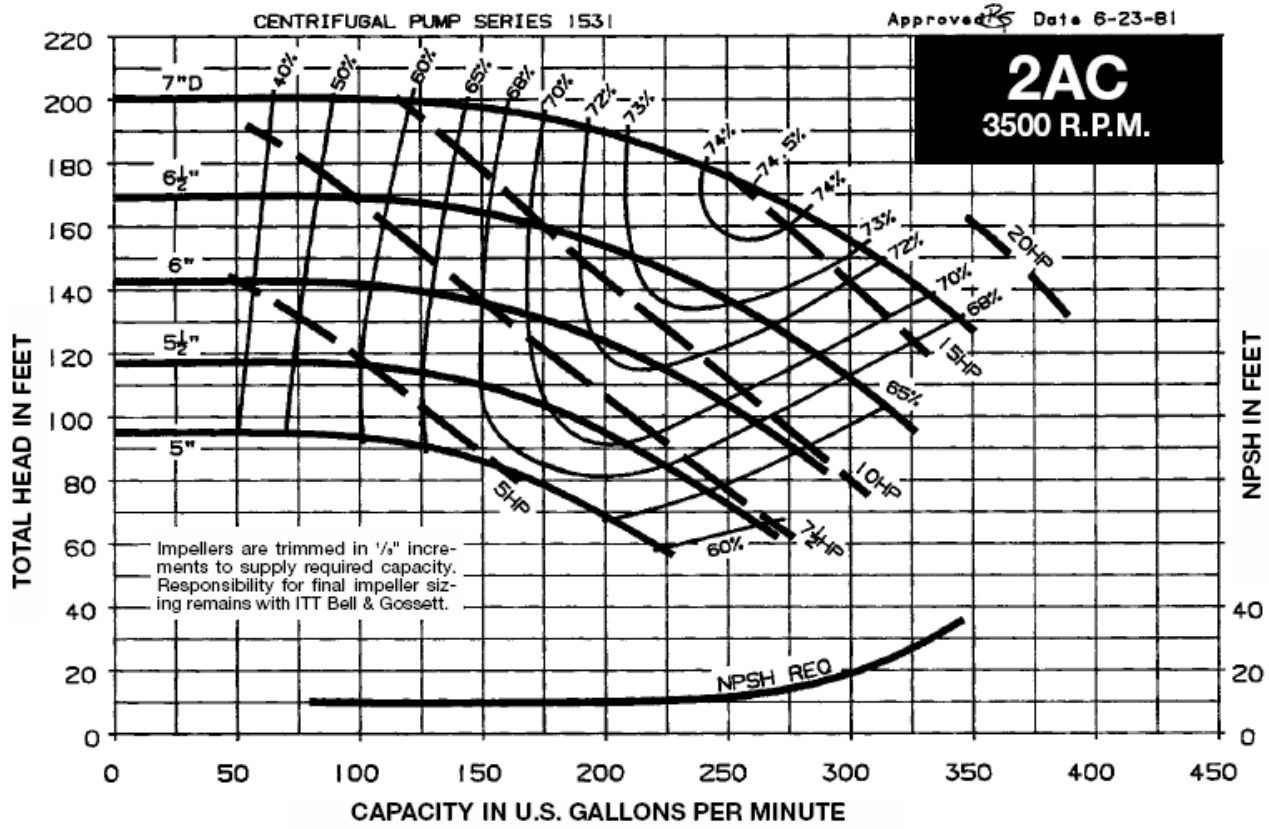
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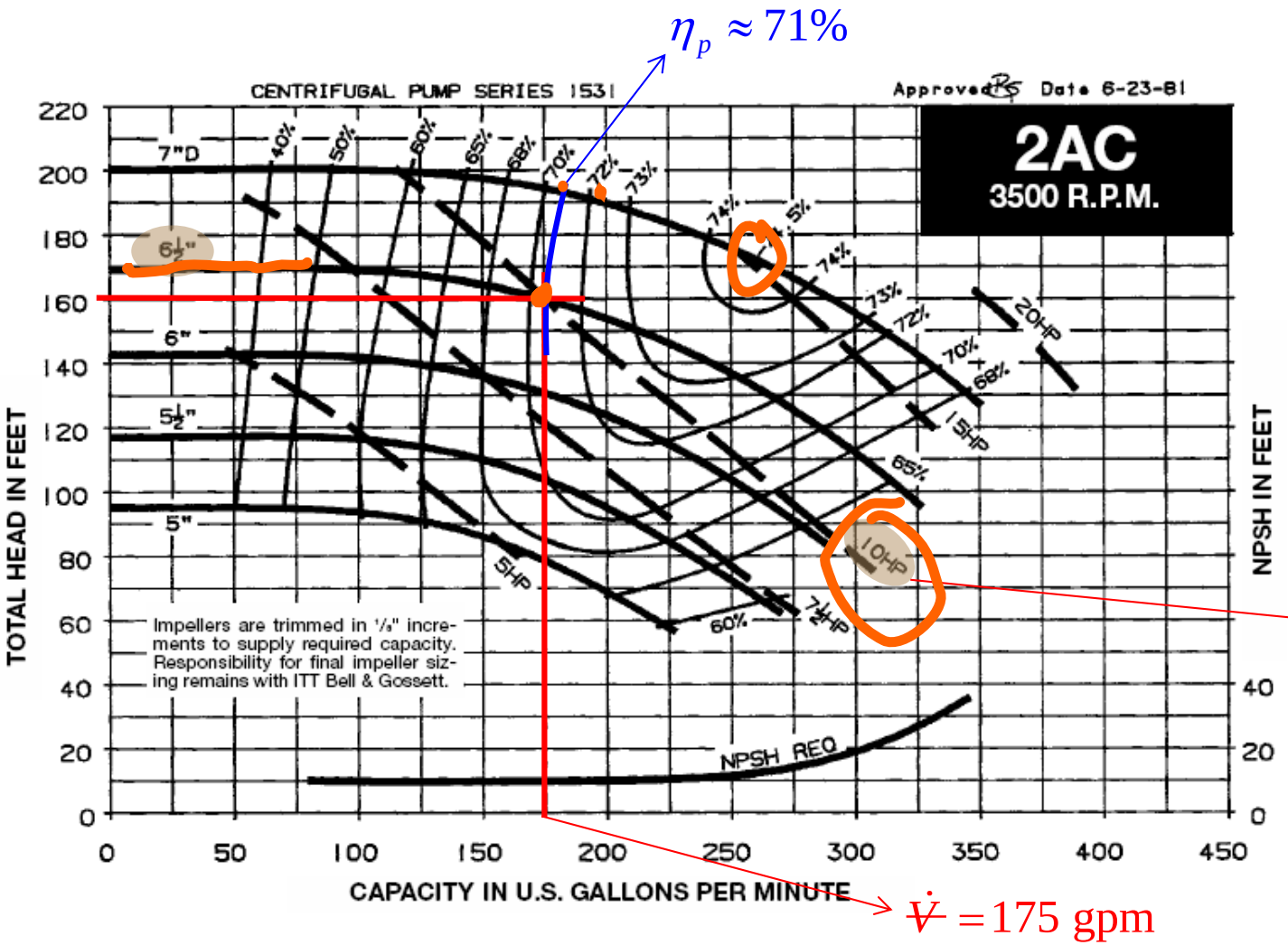


Reading and Understanding Pump Curves

For the pump model shown below, determine the capacity, **power input to the impeller**, **power input to the fluid**, and **efficiency** of a 6½-inch impeller pump if it is producing 160 ft of head. The fluid being moved is water.



Reading and Understanding Pump Curves



Reading the 6½-in impeller curve at 160 ft of head

Reading and Understanding Pump Curves

Which power is being read on the pump curve??
The power input to the fluid is given by,

$$\dot{W} = \dot{m} g H_p \qquad \dot{m} = \rho \dot{V} = \left(62.4 \frac{\text{lbm}}{\text{ft}^3}\right)(175 \text{ gpm})\left|\left(\frac{\text{ft}^3}{448.83 \text{ gpm-s}}\right) = 24.33 \frac{\text{lbm}}{\text{s}}\right.$$

$$\therefore \quad \dot{W} = \left(24.33 \frac{\text{lbm}}{\text{s}}\right)\left(32.174 \frac{\text{ft}}{\text{s}^2}\right)(160 \text{ ft})\left|\left(\frac{\text{lbf-s}^2}{32.174 \text{ lbm-ft}}\right)\left(\frac{\text{hp-s}}{550 \text{ ft-lbf}}\right) = \underline{\underline{7.08 \text{ hp}}}\right.$$

The value read from the pump curve was 10 hp. This must be the bhp input to the impeller. Checking ...

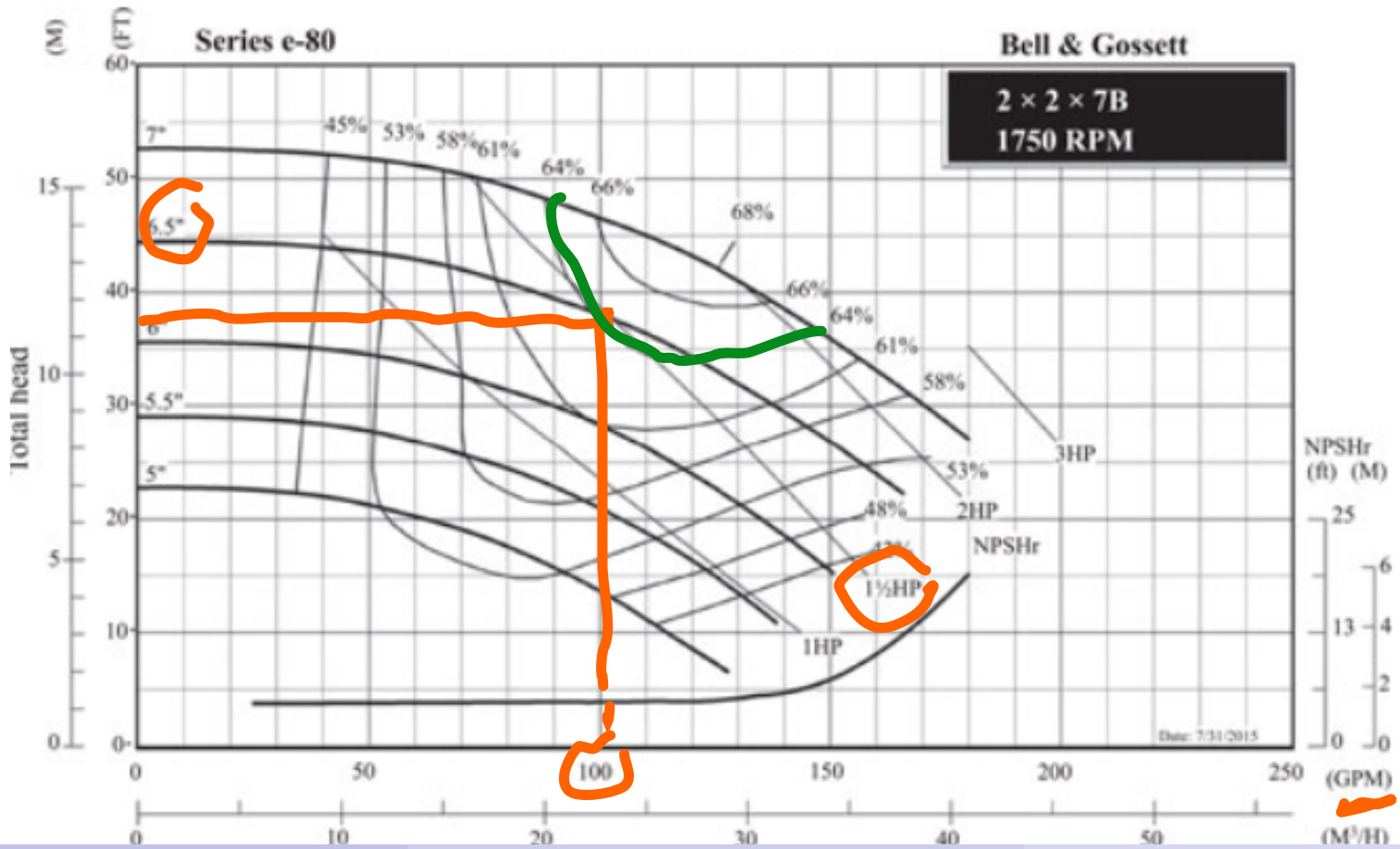
$$\eta_p = \frac{\dot{W}}{\dot{W}_{\text{bhp}}} = \frac{7.08 \text{ hp}}{10 \text{ hp}} = \underline{\underline{0.708}}$$

A pump efficiency of 71% was read from the pump curve

Example pump problem

The pump performance shown in Figure below is fitted with a 6.5-inch impeller. The pump is operating in a system and delivering 100 gpm. Determine

- the head delivered by the pump (38 ft)
- the pressure rise across the pump assuming that the fluid is water at 70°F (16.4 psi)
- the brake horsepower input to the pump (1.5 hp)
- the efficiency of the pump (64%)



System curve

The system curve is a plot of the system head as a function of volumetric flow rate. The system head is analyzed between two points in a pipe system and is made up of any pressure, velocity, and elevation heads that may exist, as well as the total head loss due to friction.

Example

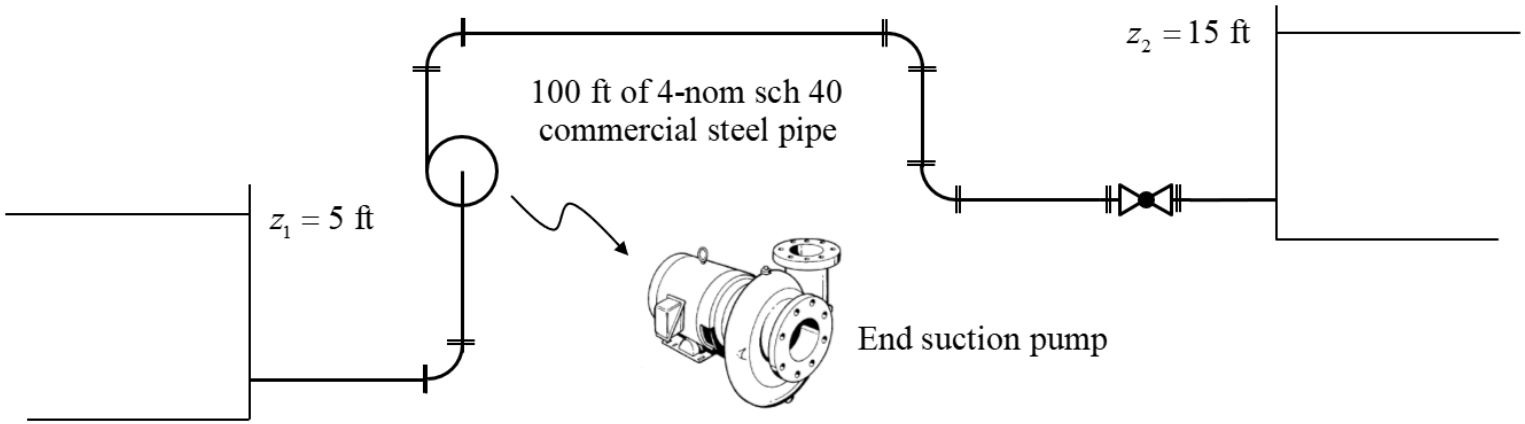
Systems are often designed to deliver a specified flow rate. For example, in a beverage bottling line where **six 12-oz bottles are filled every 10 seconds**, a fixed flow rate must be met.
If the flow rate is higher, then the **bottles will overfill**. If the flow rate is smaller, the bottles **will not fill completely**.
There are cases where a system is retrofitted which results in an increased or decreased flow rate compared to the initial design.
In cases like this, replacing the pump may not be necessary, but it is important to know how the pump will respond to changes in the system flow rate. These factors make pump selection a very important step in the complete design of a fluid delivery system. Once a pipe system has been designed, the first step toward selecting a pump is to develop what is known as **the system curve.**

System curve

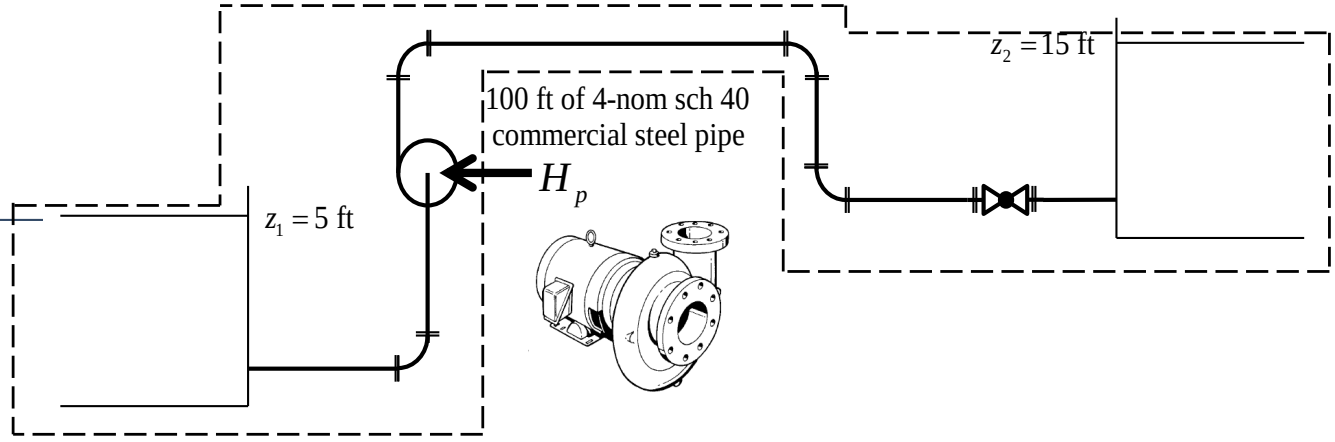
1. The pump selected for a specific application must deliver the **proper flow rate** while overcoming the elevation and friction effects in the pipe system.
2. Knowledge of how the fluid flows through the pipe system as a function of **elevation and friction effects must be fully understood**.
3. This cannot be accomplished until the **system is completely designed**.
4. The design engineer must **know the type of pipe, its diameter, total straight length, and number of fittings,** in order to perform a full energy analysis of the network to determine the elevation and friction effects.

System curve example

A piping system shown below used to transport acetone at 80°F. The system curve up to 500 gpm.
Select a pump that will provide a flow rate of 300 gpm to this system



Example



The energy equation between points 1 and 2 is ...

$$H_p + \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + z_2 + \left(f\frac{L}{D} + K_{ent} + 5K_e + K_v + K_{ex} \right)\frac{V^2}{2g}$$

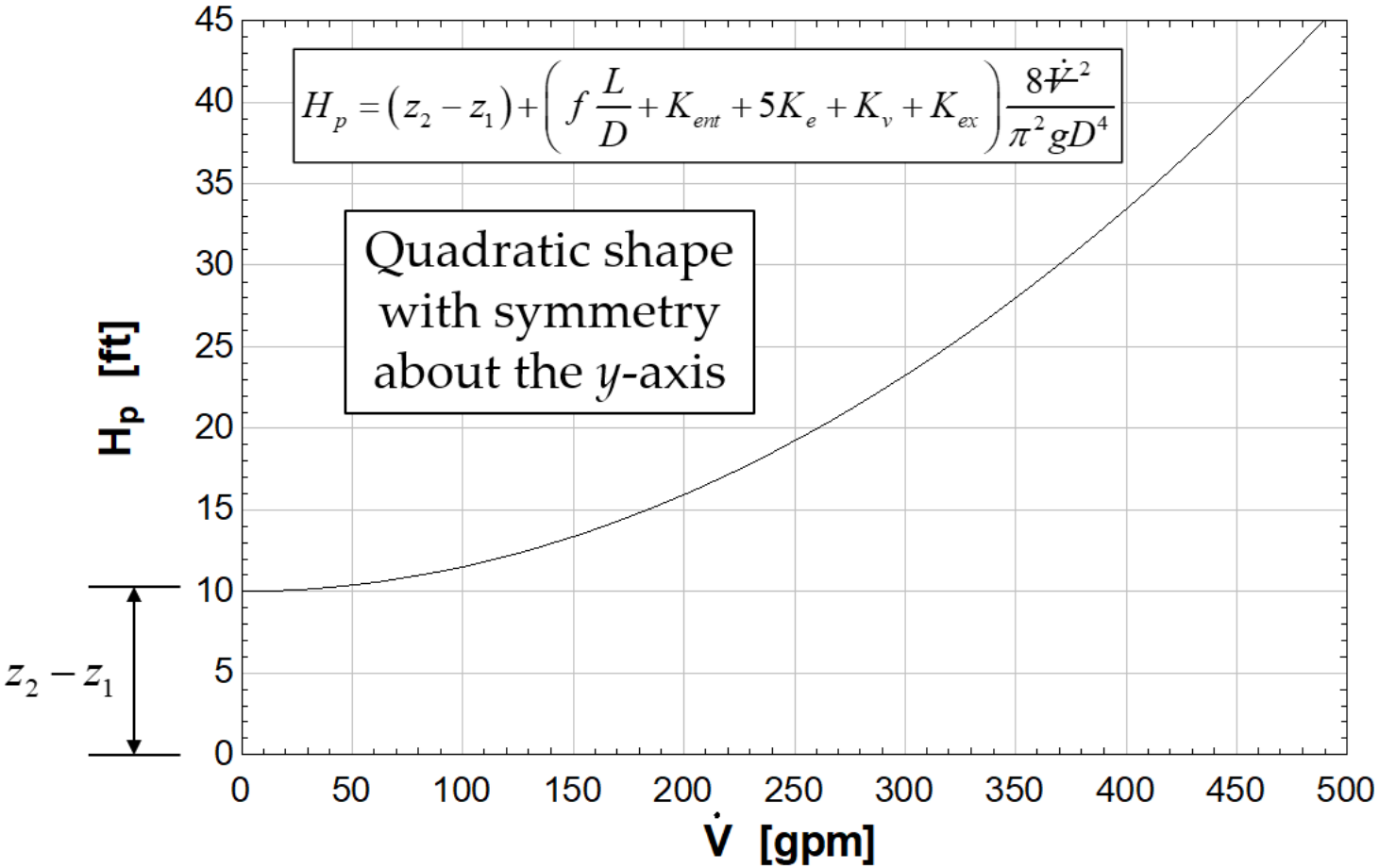
$$H_p = (z_2 - z_1) + \left(f\frac{L}{D} + K_{ent} + 5K_e + K_v + K_{ex} \right)\frac{V^2}{2g}$$

Rewriting the velocity head in terms of the volume flow rate,

$$H_p = (z_2 - z_1) + \left(f\frac{L}{D} + K_{ent} + 5K_e + K_v + K_{ex} \right)\frac{8\dot{V}^2}{\pi^2 g D^4}$$

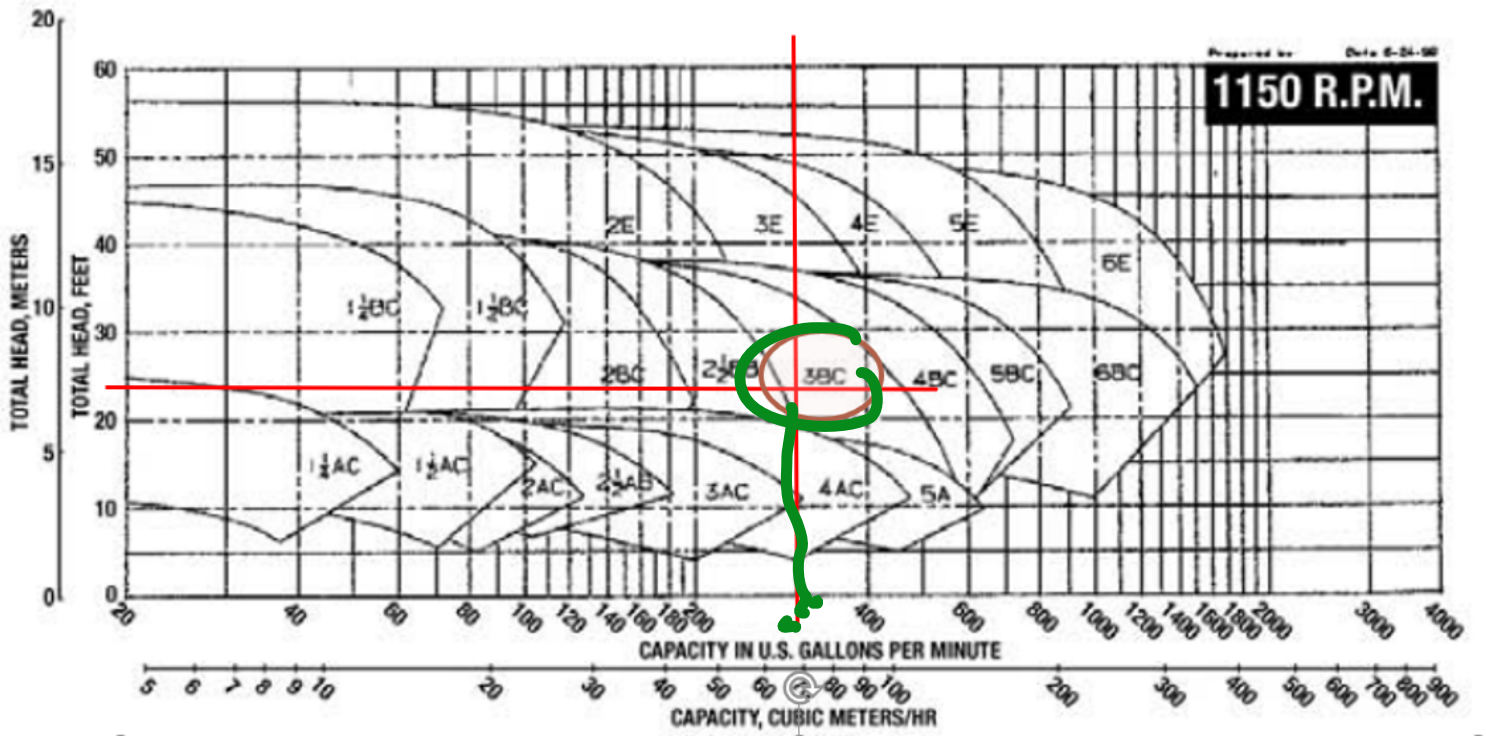
System curve example

Result of the parametric study is the **system curve**



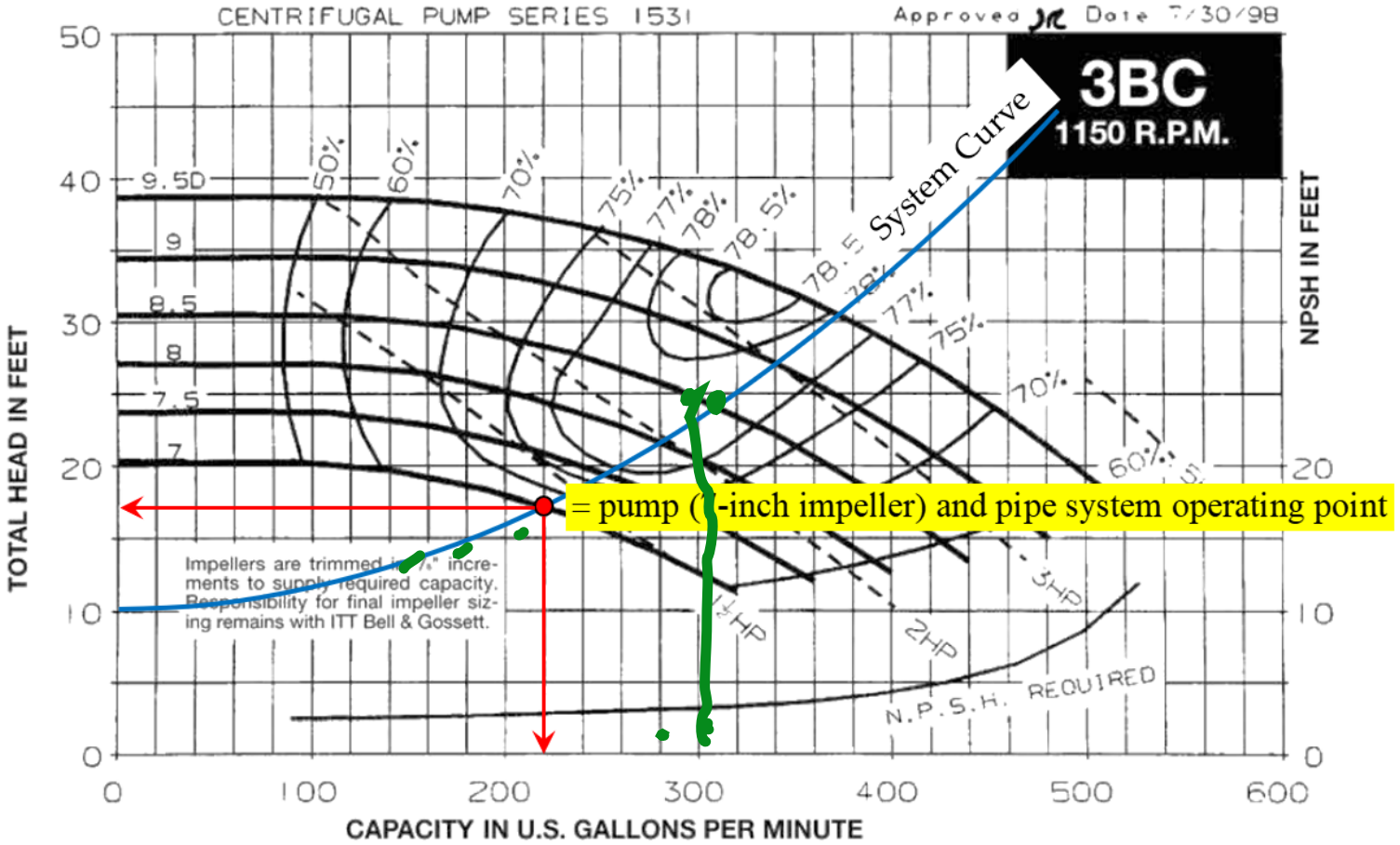
System curve example

The manufacturer's general performance map of the 1531 model pump operating at 1150 rpm is shown below.
Based on this figure, it looks like a model 3BC operating at 1150 rpm may work!



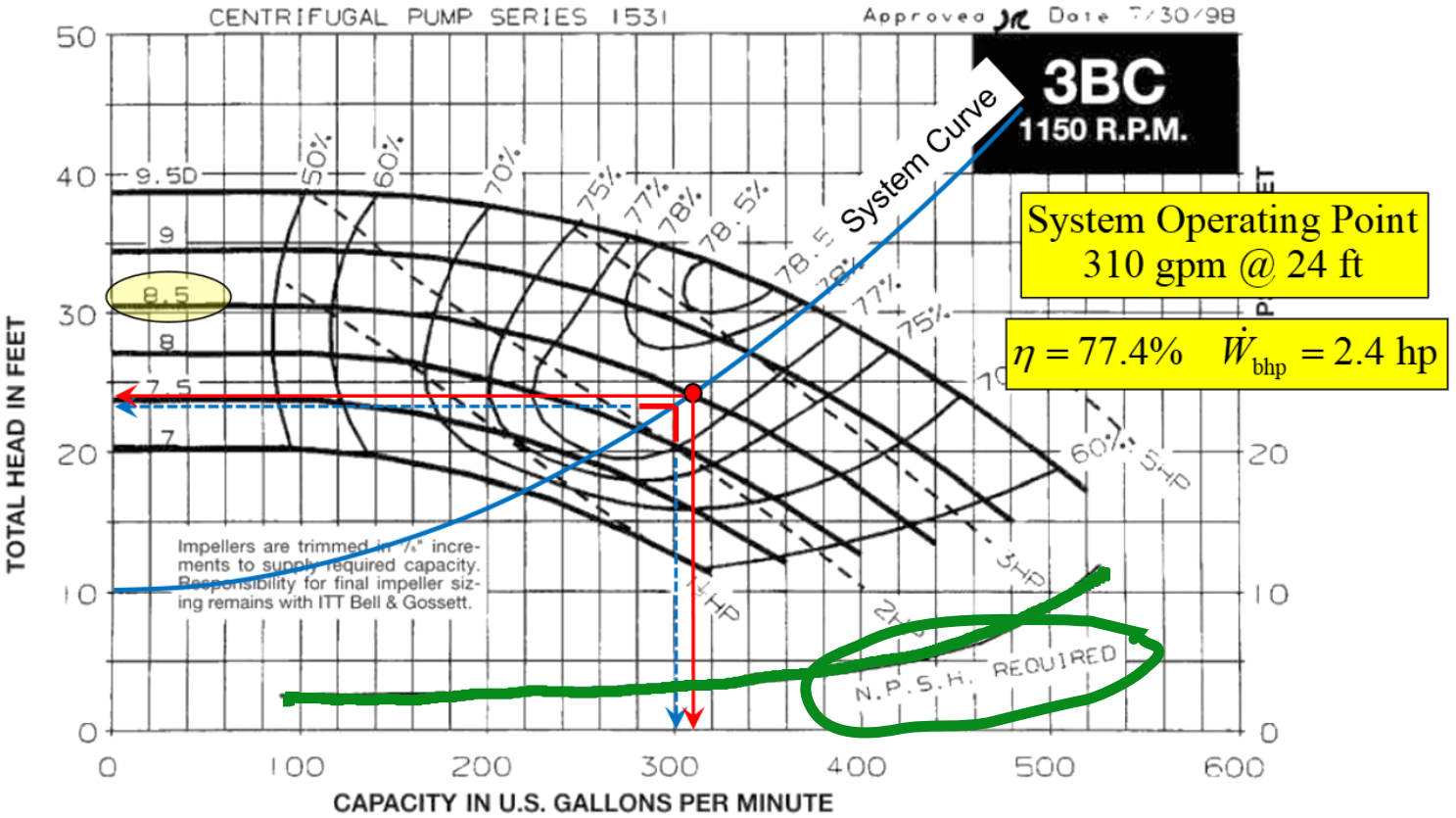
System curve example

Overlay the system curve on pump performance curve to identify pump impeller size



System curve example

Select the 8 inch impeller. System operating point 310 gpm at 24 ft head. η=77.4 and



$\dot{W}_{bhp}$=2.4 hp

System curve example

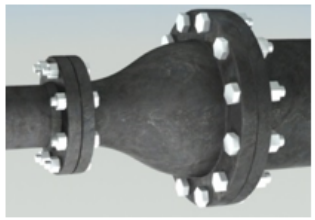
Connecting pump to existing system

The '3' indicates 3-nom suction and discharge ports on the pump. Since the pipeline is 4-nom, a reducer and expander are needed to connect it to the pipeline.



reducer

3BC Pump

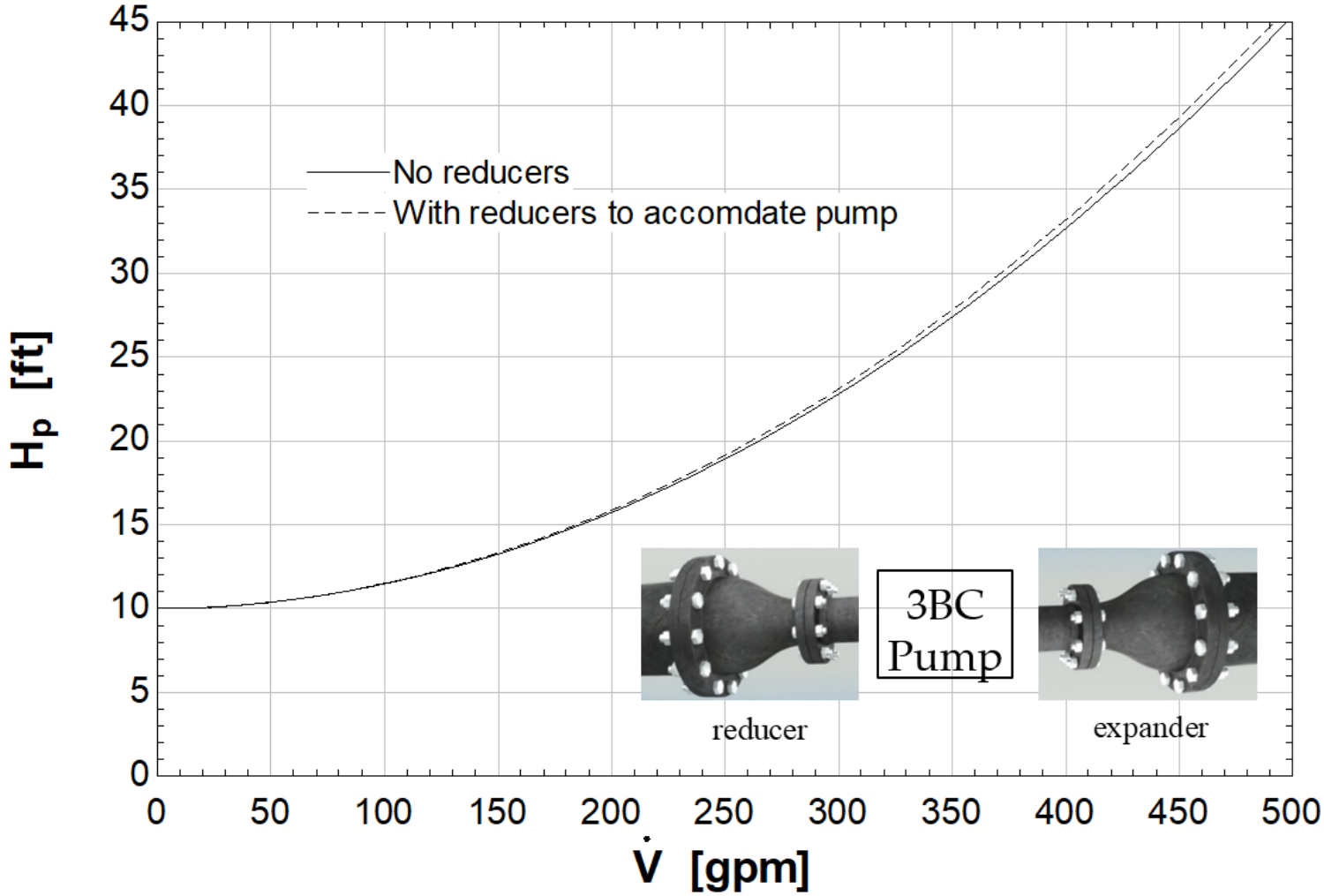


expander

The system curve must be recalculated to include the new resistances introduced by the reducer and expander (K-values computed using minor losses table).

System curve example

The added resistance moves the system curve to the left. In this case, the additional resistance is negligible!

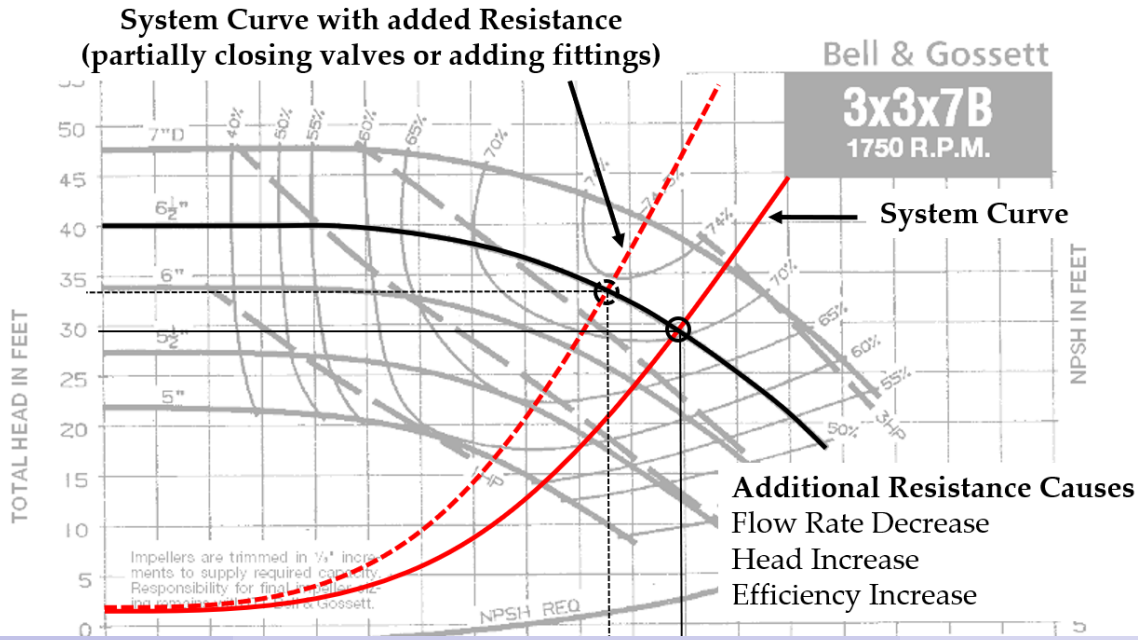


System curve example

Final selection summary

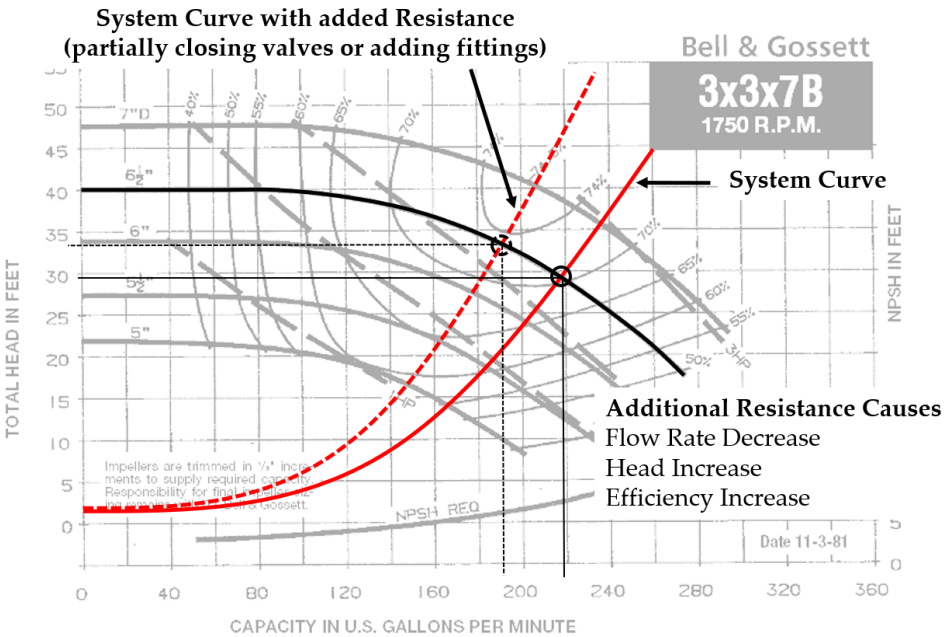
- Bell & Gossett Series 1531 End Suction - Model 3BC
  - reducer and expander required to connect to pipeline
- Operating speed: 1150 rpm
- Operating point
  - 310 gpm at 24 ft
  - Efficiency: 77.4
  - Brake horsepower input: 2.4 hp

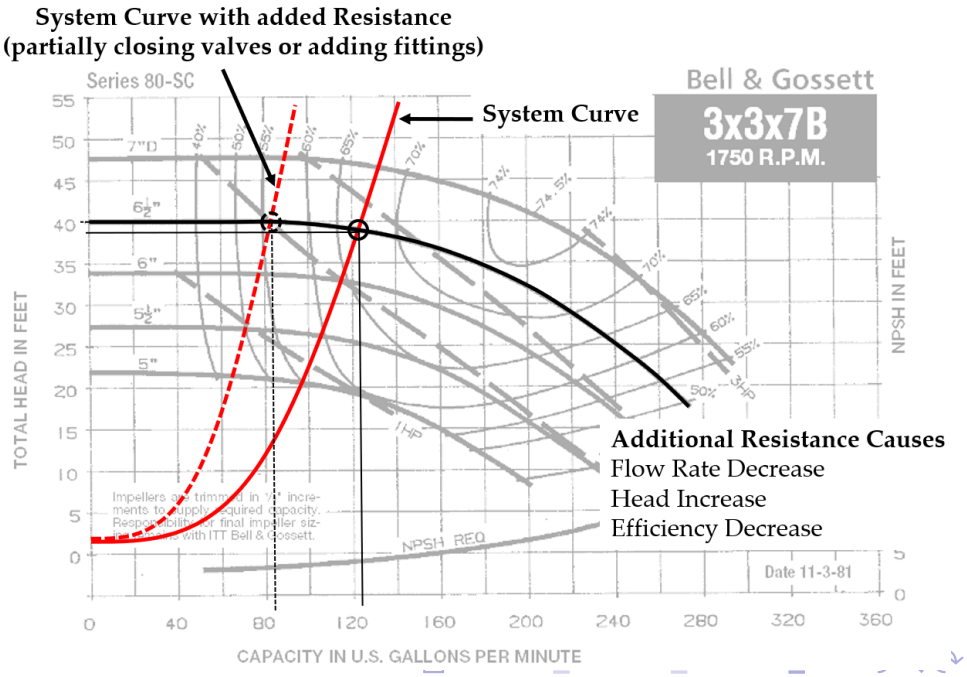
The flow can be controlled with the globe valve.
Partially closing the valve increases the pump head which increases the bhp input to the pump In this application, the efficiency of the pump experiences a slight increase



General observations

- The system curve must be determined in order to properly size a pump
- The operating point of the pump/pipe network is the intersection of the pump performance curve and the system curve
- Additional resistances may need to be considered if the suction and discharge ports of the pump are different than the pipe
  - The system curve must be recalculated and a new operating point can be determined
- Additional resistance in the pipe network (e.g., closing a valve) effectively shift the system curve to the left
  - This changes the operating point of the pump/pipe system including the pump efficiency and the bhp input to the pump



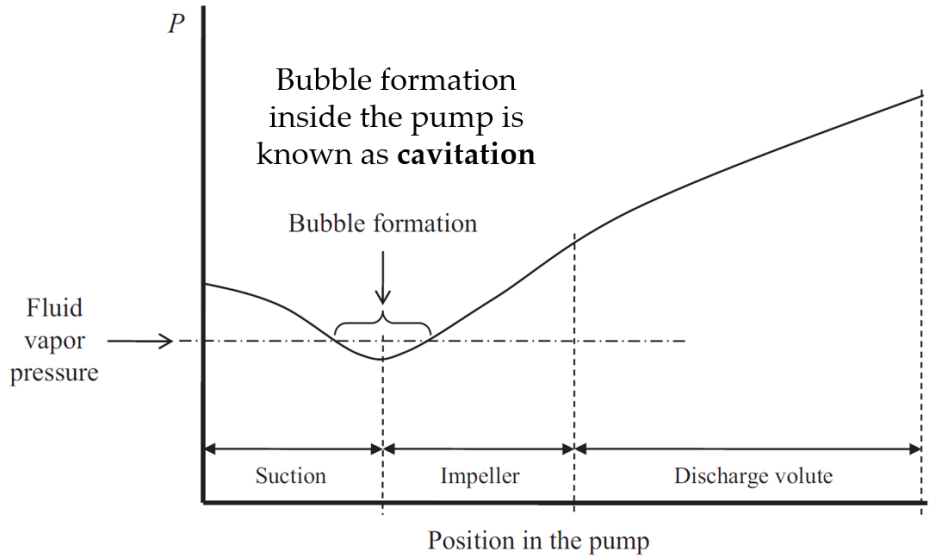


General observations

- Pumps should be selected to operate slightly to the right of the maximum efficiency
  - Unforeseen system resistances (e.g., partially closing valves for flow control, and/or adding additional valves or fittings) will cause head increase, capacity decrease, but an efficiency increase
- Selecting pumps that operate to the left of the maximum efficiency may result in motor failure if the efficiency drops too much due to unforeseen flow resistances

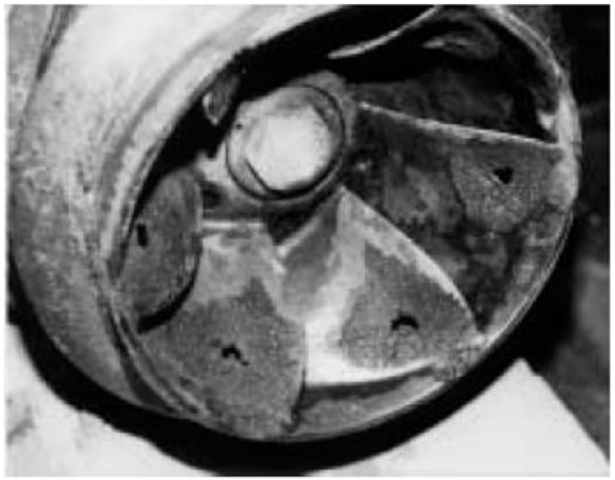
Cavitation

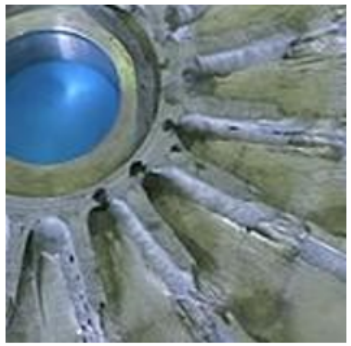
Pressure trace inside the pump



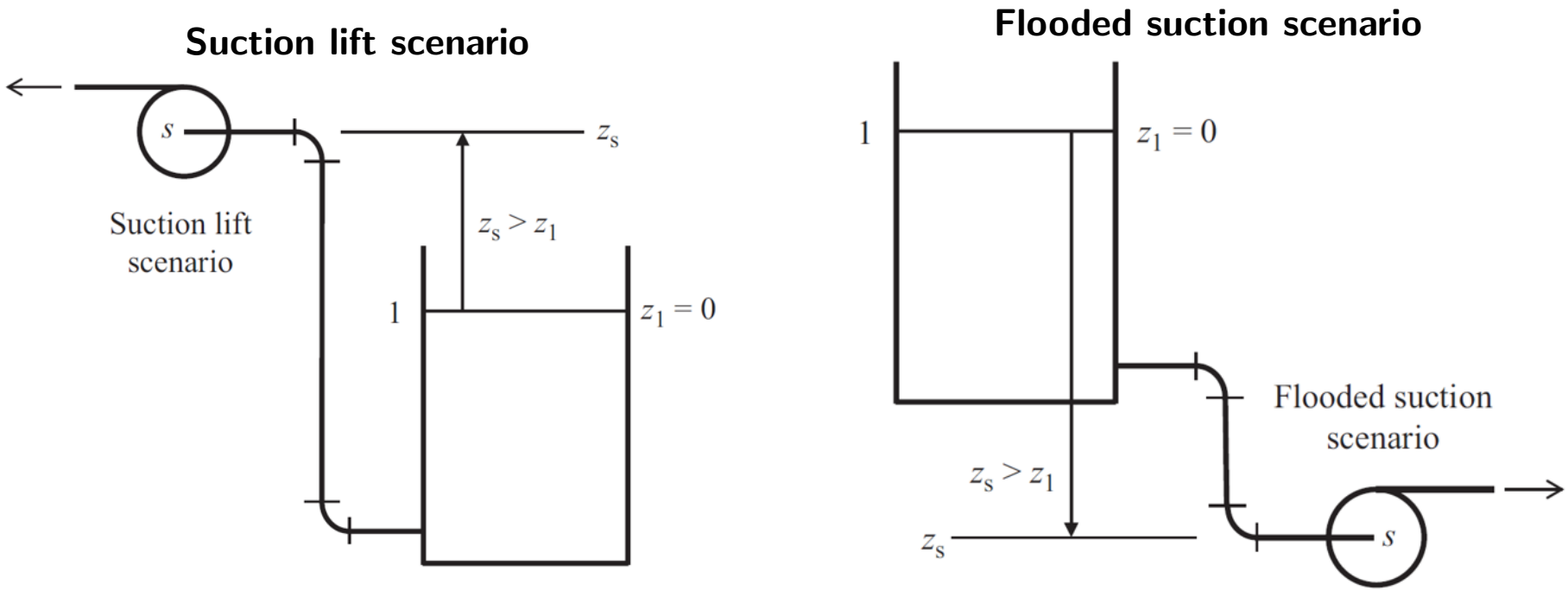
Why cavitation is detrimental to pumps?

- Bubble collapse due to high pressure causes pressure waves to propagate
- Pressure waves cause cavitation erosion and possible catastrophic failure of the pump





Net Positive Suction Head



Solving the Conservation of Energy equation for the pressure at the suction port of the pump results in,

$$\frac{P_s}{\gamma} = \frac{P_1}{\gamma} + (z_1 - z_s) - \left(f\frac{L}{D} + \sum K + 1 \right) \frac{V_s^2}{2g} \tag{8}$$

Net Positive Suction Head

NPSHa

If Ps = Pv (vapor pressure), then cavitation will definitely occur. The difference between Ps and Pv represents the suction head available in the suction line of the piping system. In head terms, this difference is known as the **Net Positive Suction Head Available (NPSHa)**.

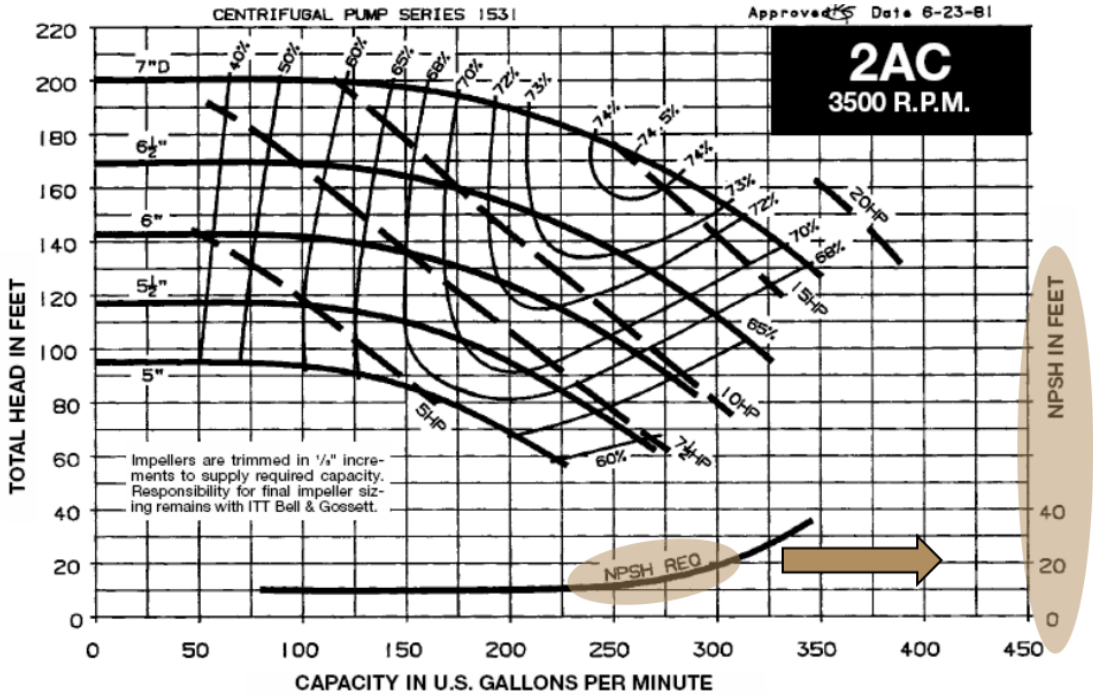
$$\text{NPSHa} = \frac{P_s}{\gamma} - \frac{P_v}{\gamma} \tag{9}$$

$$\text{NPSHa} = \frac{P_1}{\gamma} + (z_1 - z_s) - \left(f\frac{L}{D} + \sum K + 1\right)\frac{V_s^2}{2g} - \frac{P_v}{\gamma} \tag{10}$$

- $(z_1 - Z_2)$
  - Negative for suction lift systems
  - Positive for flooded suction systems
  - Lift system are more prone to cavitation
- $NPSHa \leq 0$
  - The fluid is boiling at the pump inlet
  - Cavitation will occur
- $NPSHa \geq 0$
  - No boiling at inlet
  - Cavitation still may occur due to ΔP before impeller

Net Positive Suction Head

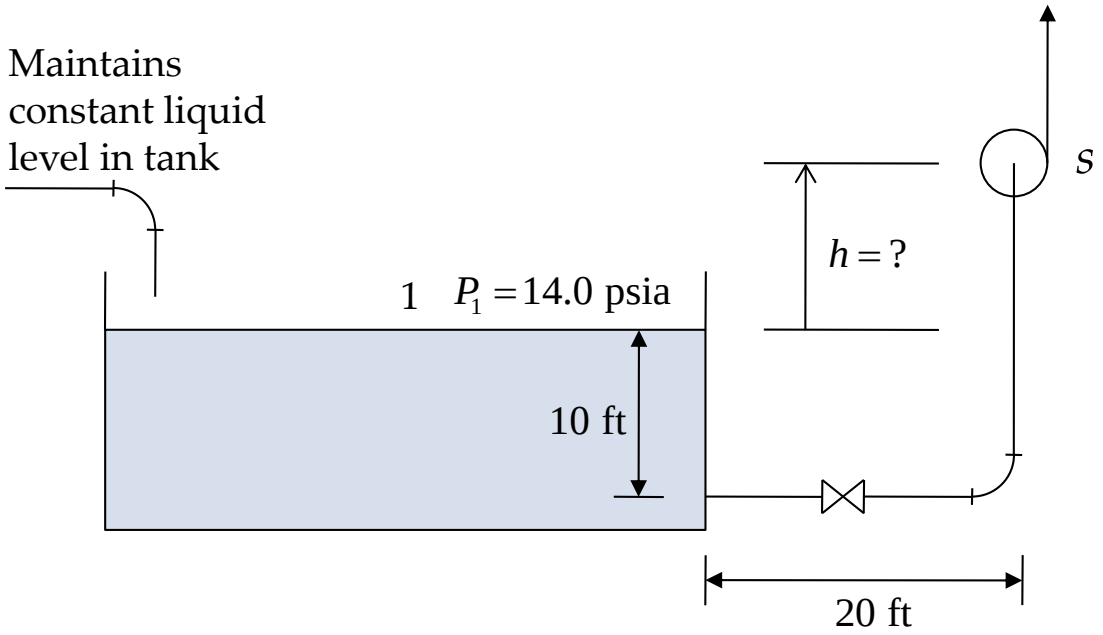
NPSHa > 0 does not guarantee cavitation prevention. How can we be assured that cavitation will not occur?

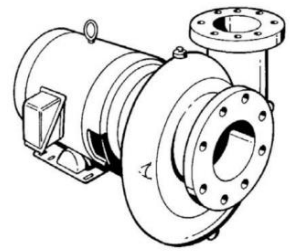


Manufacturers of pumps determine the Net Positive Suction Head Required (NPSHr) to prevent cavitation. Therefore, cavitation will be prevented when, NPSHa $>$ NPSHr
Recommended saftey factor NPSHa-NPSHr ≥ 3 ft

Pump Placement to Prevent Cavitation

Given: Water at 100°F is being pumped vertically upward from a holding tank by a Bell & Gossett Series 1531 Model 2AC 6½-inch impeller pump operating at 3500 rpm. When the pump is installed in the pipe system, the head produced by the pump is 150 ft.

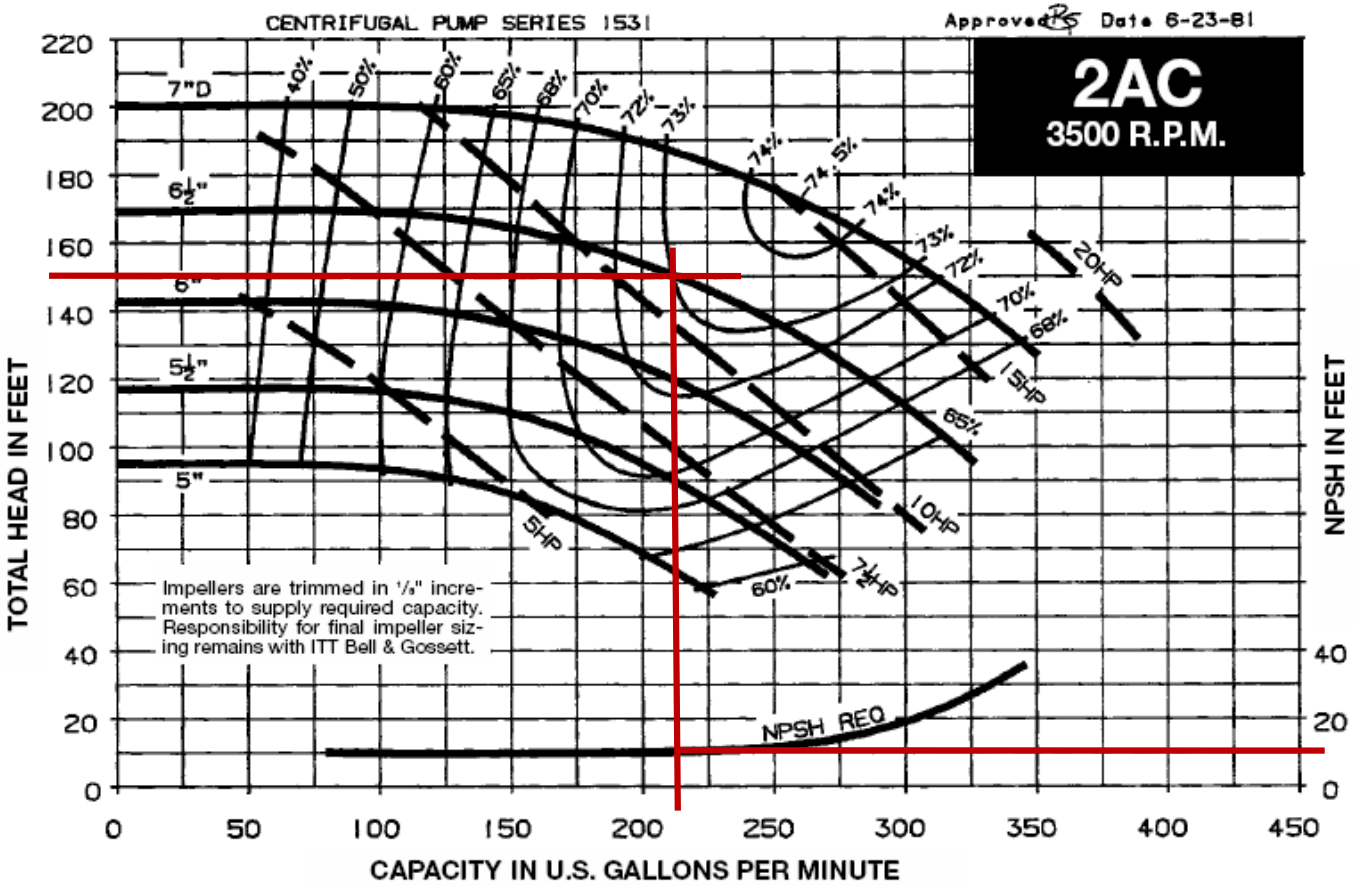




3-nom sch 40
commercial steel pipe

Find: Where should the pump be placed in the suction line to prevent cavitation?

Pump Placement to Prevent Cavitation



From the pump curve

$\dot{V} = 212.5$ gpm

NPSHr = 10 ft

Pump Placement to Prevent Cavitation

The NPSHa is given by,

$$\text{NPSHa} = \frac{P_s}{\gamma} - \frac{P_v}{\gamma} = \frac{P_1}{\gamma} + \left(z_1 - z_s\right) - \left(f\frac{L}{D} + \sum K + 1 \right)\frac{V_s^2}{2g} - \frac{P_v}{\gamma}$$

This NPSHa must exceed the NPSHr by at least 3 ft.

$$\left[\frac{P_1}{\gamma} + \left(z_1 - z_s\right) - \left(f\frac{L}{D} + \sum K + 1 \right)\frac{V_s^2}{2g} - \frac{P_v}{\gamma}\right] - \text{NPSHr} = 3 \text{ ft}$$

We are interested in the elevation difference that makes this true,

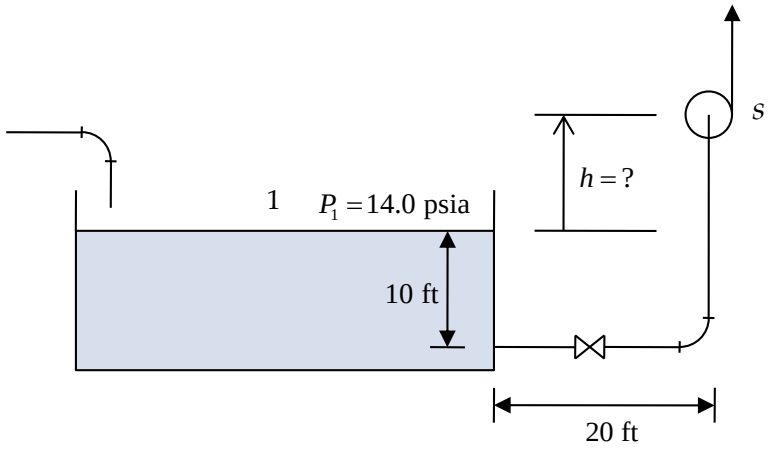
$$h = \left(z_s - z_1\right) = \frac{P_1}{\gamma} - \left(f\frac{L}{D} + \sum K + 1 \right)\frac{V_s^2}{2g} - \frac{P_v}{\gamma} - \text{NPSHr} - 3 \text{ ft}$$

Pump Placement to Prevent Cavitation

$$h = (z_s - z_1) = \frac{P_1}{\gamma} - \left(f\frac{L}{D} + \sum K + 1 \right)\frac{V_s^2}{2g} - \frac{P_v}{\gamma} - \text{NPSHr} - 3 \text{ ft}$$

Total straight pipe length is not known yet! Referring to the system sketch,

$$L = (20 + 10 + h) \text{ ft}$$



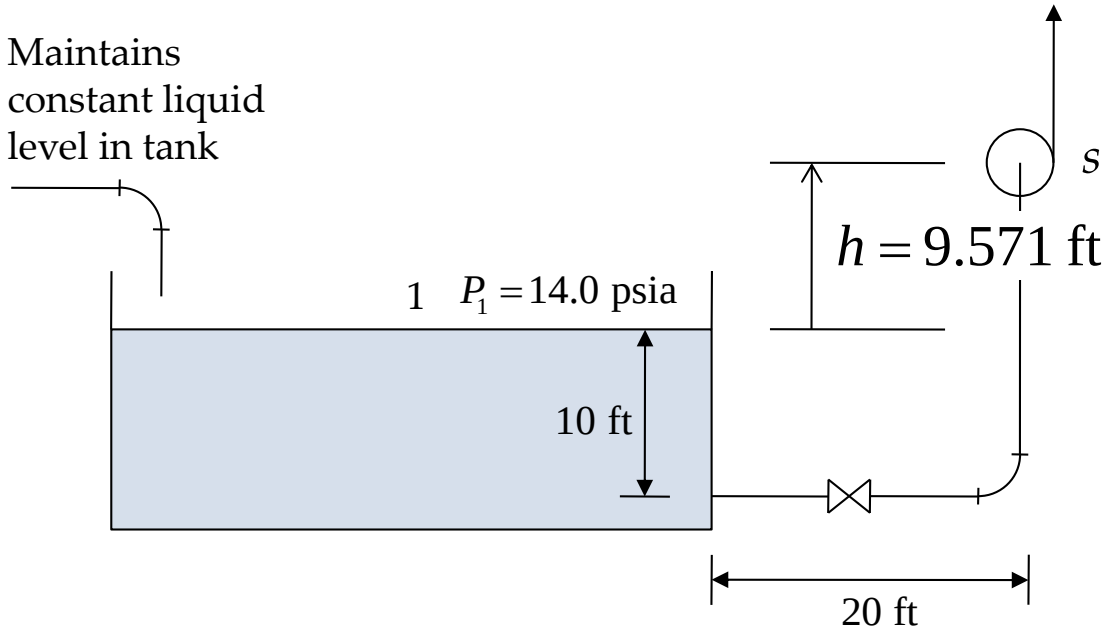
Therefore,

$$h = \frac{P_1}{\gamma} - \left[f\frac{(30[\text{ft}] + h)}{D} + \sum K + 1 \right]\frac{V^2}{2g} - \frac{P_v}{\gamma} - \text{NPSHr} - 3 \text{ ft}$$

Pump Placement to Prevent Cavitation

$$h = \frac{P_1}{\gamma} - \left[f \frac{\left(30[\text{ft}] + h\right)}{D} + \sum K + 1 \right] \frac{V^2}{2g} - \frac{P_v}{\gamma} - \text{NPSHr} - 3 \text{ ft}$$

Solving for h,



Observations

- The elevation computed in the example problem represents the **maximum** height above the reservoir surface that the pump should be placed
  - Any elevation higher may cause cavitation to occur due to the extra suction head loss
- Equations developed here and in the text are for pumping a liquid from a large tank
  - Other applications require a full energy analysis to determine the NPSHa expression

Dimensional analysis and similitude

Similitude

If all pumps of one type (e.g., centrifugal) behave similarly, then dimensional analysis should reveal how pump parameters scale.

Dimensional analysis

Dimensional analysis helps us determine which dimensionless groups of variables fully describe the operation of similar pumps.

$$\eta = f_1\left(\frac{\rho\omega D^2}{\mu}, \frac{\dot{V}}{\omega D^3}\right) \tag{11}$$

$$\frac{gH_p}{\omega^2 D^2} = f_2\left(\frac{\rho\omega D^2}{\mu}, \frac{\dot{V}}{\omega D^3}\right) \tag{12}$$

$$\frac{\dot{W}}{\rho\omega^3 D^5} = f_3\left(\frac{\rho\omega D^2}{\mu}, \frac{\dot{V}}{\omega D^3}\right) \tag{13}$$

$$\frac{\rho\omega D^2}{\mu} = \text{Rotational Reynolds number} \tag{14}$$

$$\frac{\dot{V}}{\omega D^3} = \text{Volumetric flow coefficient} \tag{15}$$

Dimensional analysis and similitude

Experiments have shown that the efficiency, head coefficient, and power coefficient are only weakly dependent on the Rotational Reynolds Number. Therefore,

$$\eta \simeq f_1\left(\frac{\dot{\forall}}{\omega D^3}\right) \tag{16}$$

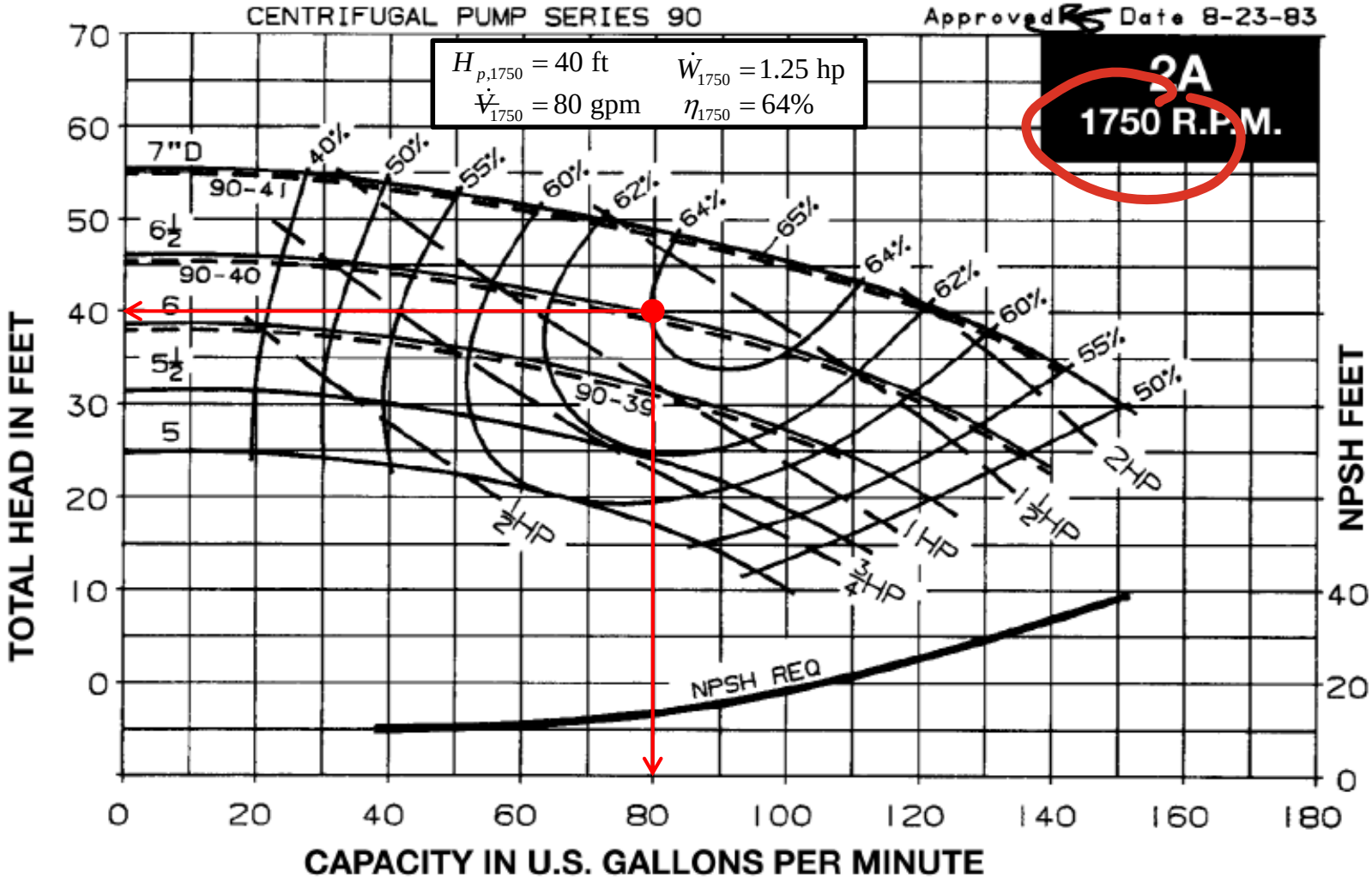
$$\frac{gH_p}{\omega^2 D^2} \simeq f_2\left(\frac{\dot{\forall}}{\omega D^3}\right) \tag{17}$$

$$\frac{\dot{W}}{\rho\omega^3 D^5} \simeq f_3\left(\frac{\dot{\forall}}{\omega D^3}\right) \tag{18}$$

(19)

Performance predictions for a pump can be made based on the known performance of a geometrically similar pump.
These equations are known as the Affinity Laws or Similarity Laws

Application of the Affinity Laws



Application of the Affinity Laws

For the 6½-inch impeller at 1750 rpm,

$$\Delta H_{1750} = 40 \text{ ft} \qquad \dot{W}_{1750} = 1.25 \text{ hp}$$
$$\dot{\forall}_{1750} = 80 \text{ gpm} \qquad \eta_{1750} = 64\%$$

What is the **estimated** performance of this pump when the rotational speed is changed to 3500 rpm?

$$\eta \approx f_1\left(\frac{\dot{\forall}}{\omega D^3}\right)$$
$$\frac{g\Delta H}{\omega^2 D^2} \approx f_2\left(\frac{\dot{\forall}}{\omega D^3}\right)$$
$$\frac{\dot{W}}{\rho\omega^3 D^5} \approx f_3\left(\frac{\dot{\forall}}{\omega D^3}\right)$$

Capacity – Equate the Flow Coefficients

$$\frac{\dot{\forall}_{3500}}{\omega_{3500} D_{3500}^3} = \frac{\dot{\forall}_{1750}}{\omega_{1750} D_{1750}^3}$$

$$\dot{\forall}_{3500} = \dot{\forall}_{1750}\frac{\omega_{3500}}{\omega_{1750}} = (80 \text{ gpm})\left(\frac{3500 \text{ rpm}}{1750 \text{ rpm}}\right) = \underline{\underline{160 \text{ gpm}}}$$

Application of the Affinity Laws

$$\eta \approx f_1\left(\frac{\dot{\forall}}{\omega D^3}\right)$$

$$\frac{g\Delta H}{\omega^2 D^2} \approx f_2\left(\frac{\dot{\forall}}{\omega D^3}\right)$$

$$\frac{\dot{W}}{\rho\omega^3 D^5} \approx f_3\left(\frac{\dot{\forall}}{\omega D^3}\right)$$

Head – Equate the Head Coefficients

$$\frac{gH_{p,3500}}{\omega_{3500}^2 D^2} = \frac{gH_{p,1750}}{\omega_{1750}^2 D^2}$$

$$H_{p,3500} = H_{p,1750}\left(\frac{\omega_{3500}}{\omega_{1750}}\right)^2 = (40\text{ ft})\left(\frac{3500\text{ rpm}}{1750\text{ rpm}}\right)^2 = \underline{\underline{160\text{ ft}}}$$

Power – Equate the Power Coefficients

$$\frac{\dot{W}_{3500}}{\rho\omega_{3500}^3 D^5} = \frac{\dot{W}_{1750}}{\rho\omega_{1750}^3 D^5}$$

$$\dot{W}_{3500} = \dot{W}_{1750}\left(\frac{\omega_{3500}}{\omega_{1750}}\right)^3 = (1.25\text{ hp})\left(\frac{3500\text{ rpm}}{1750\text{ rpm}}\right)^3 = \underline{\underline{10\text{ hp}}} \quad \leftarrow$$

Application of the Affinity Laws

Efficiency – Equate the Efficiencies

The efficiency is a dimensionless group. As with the other dimensionless groups we expect that,

$$\eta_{3500} = \eta_{1750} = 64\%$$

$$\eta \approx f_1\left(\frac{\dot{\forall}}{\omega D^3}\right)$$

$$\frac{g\Delta H}{\omega^2 D^2} \approx f_2\left(\frac{\dot{\forall}}{\omega D^3}\right)$$

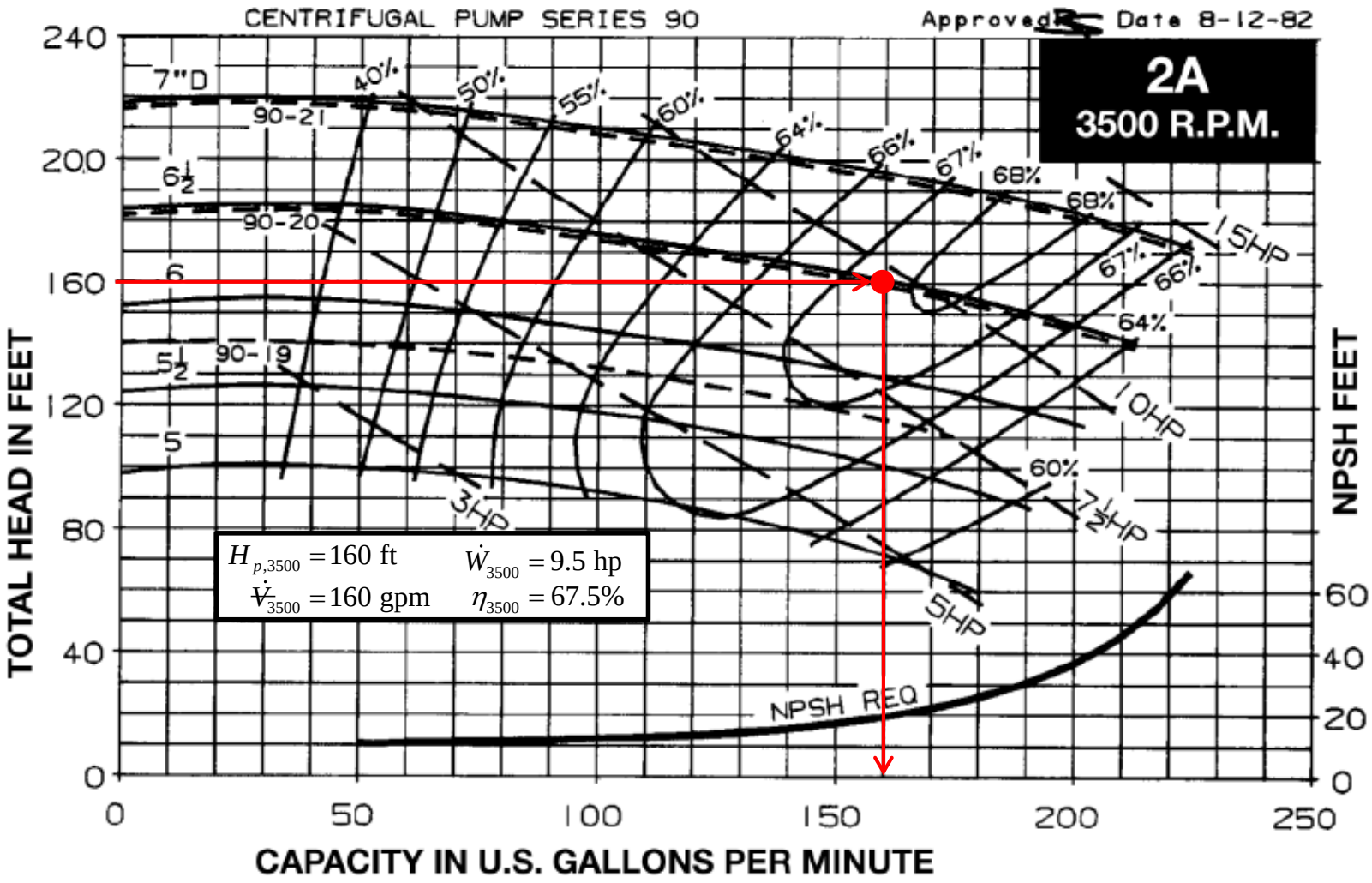
$$\frac{\dot{W}}{\rho\omega^3 D^5} \approx f_3\left(\frac{\dot{\forall}}{\omega D^3}\right)$$

For a constant diameter impeller where only the speed of the pump is being varied, the Affinity Laws can be stated as,

$$\frac{\dot{\forall}_n}{\dot{\forall}_o} = \frac{\omega_n}{\omega_o} \qquad \frac{H_{p,n}}{H_{p,o}} = \left(\frac{\omega_n}{\omega_o}\right)^2 \qquad \frac{\dot{W}_n}{\dot{W}_o} = \left(\frac{\omega_n}{\omega_o}\right)^3 \qquad \eta_n = \eta_o$$

How accurate are these expressions?

Application of the Affinity Laws



Summary and Observations

Parameter	1750 rpm (Pump Curve)	3500 rpm (Affinity Laws)	3500 rpm (Pump Curve)
Capacity	80 gpm	160 gpm	160 gpm
Head Increase	40 ft	160 ft	160 ft
Power Draw	1.25 hp	10 hp	9.5 hp
Efficiency	64%	64%	67.5%

Recall ...

The Affinity Laws have been developed assuming that the dimensionless groups are only weakly dependent on the Rotational Reynolds Number. The discrepancies in the predicted power and efficiency from the Affinity Laws compared to manufacturer's data in this example demonstrates that this assumption may not be entirely correct; but close!

Even with these small discrepancies, the Affinity Laws are helpful in estimating the performance of a pump at different operating speeds!

Dimensional analysis and similitude- Example

For a constant diameter impeller where only the speed of the pump is being varied, the Affinity Laws can be stated as,

$$\frac{\dot{\forall}_n}{\dot{\forall}_o} = \frac{\omega_n}{\omega_o} \tag{20}$$

$$\frac{H_{p,n}}{H_{p,o}} = \left(\frac{\omega_n}{\omega_o}\right)^2 \tag{21}$$

$$\frac{\dot{W}_n}{\dot{W}_o} = \left(\frac{\omega_n}{\omega_o}\right)^3 \tag{22}$$

$$\eta_n = \eta_o \tag{23}$$

Pump design practice

Pump design guidelines

Design Practices

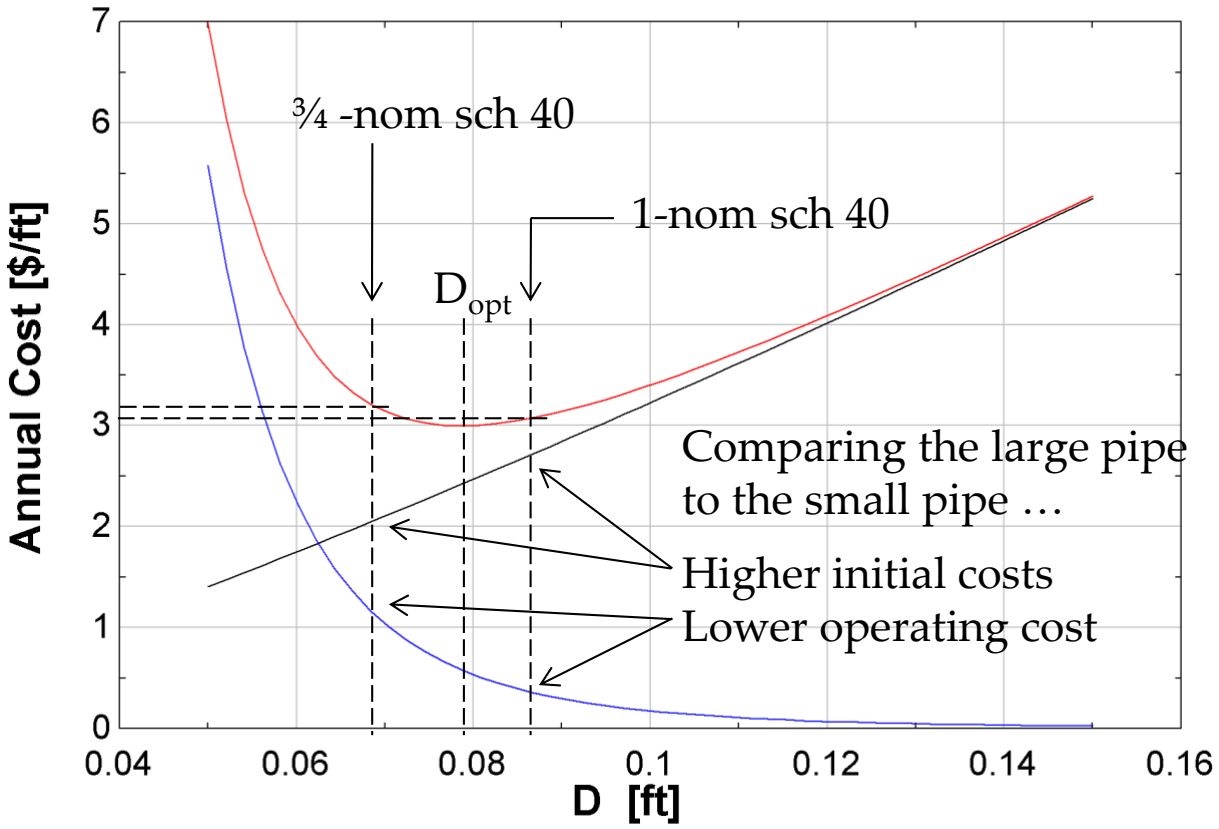
- Economic Issues
  - Wise network layout
    - Minimize capital cost and friction losses
  - Pipe system should be designed to be close to the economic diameter



Design Practices

Minimum Cost System …

In the example below, both ¾ -nom and 1-nom are very close to D_{opt}



The larger pipe size may provide more design flexibility

- Can accommodate future flow rate increases
- Lower operating cost (power costs generally do not decrease)
- Smaller pressure drop (noise and vibration suppression)
- Fouling can occur, but the effect is not as profound as in a smaller pipe
- Other issues?

Design Practices

- Environmental Impact
  - Most pumps are driven by electric motors
    - EMDS – electric motor drive systems
    - Where does the electricity originate?
  - EMDS in 2011
    - 43-46% of all global electrical energy use
    - End users of EMDS spent $565B globally
  - EMDS projected by 2030
    - 13,360 TWh (value ~ $900B)
    - 8570 Mt (megatons) of CO_2 emissions from fossil fuel power plants

Design Practices

- Pressure drop limitations to avoid noise and vibration
  - Liquids: 25-30 psi per 1000 ft
  - Gases: 10-15 psi per 1000 ft
  - Low Pressure Steam: 4 psi per 100 ft

Design Practices

- Pump placement
  - The pump must be placed to avoid cavitation
    - NPSHA analysis should be considered; especially in the case of a suction lift scenario
- Flow control
  - Single speed motors turn at a constant rpm
    - Valves need to be used for flow control (globe valves are generally used for this)
  - Variable speed motors
    - More expensive, but prices are coming down!
    - Affinity Laws can help determine the effect of varying the pump rotational speed

Design Practices

- Valve placement for flow control
  - If possible, valves should be avoided for flow control
    - Increased flow resistance
  - If a valve is necessary, it should be placed on the discharge line of the pump
    - Valves in the suction line can cause cavitation
- Use of valves
  - Flow control
  - System isolation for repair and maintenance
  - Valve selection
    - Table 4.5 and other handbooks

Design Practices

- Entrained gases in liquid systems
  - Provide a slight elevation increase in the flow direction
    - Causes gas to move to a high point in the system
  - Bleed ports (valves) or expansion tanks can be used for gas removal at strategic points in the system
- The previous slides are just a few common design practices
  - Other sources
    - Piping handbooks and guides
      - ASME, ANSI
    - Company practices
    - Experienced design engineers
    - Publications
      - Piping journals, ME Magazine, etc.

Example-1: Pump

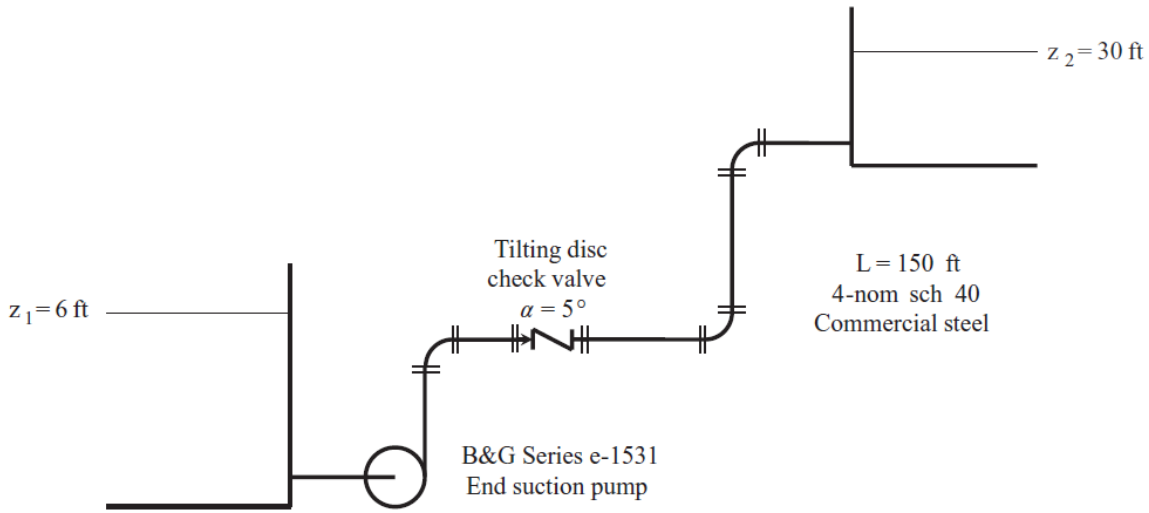
A centrifugal pump is being used to transport 45 m^3/h of a 20% ethylene glycol solution at 0°C. The suction pipe inlet to the pump is $2\frac{1}{2}$-nom sch 40 and the pump discharge pipe is 2-nom sch 40. Pressure gauges on the suction and discharge ports of the pump indicate -15 kPa and 250 kPa, respectively. The elevation difference between the suction and discharge ports of the pump is 0.08 m. At this condition, the pump has an efficiency of 68%. Determine the following:

1. The head produced by this pump (m)
2. The percentage of the total pump head that is used to increase the fluid's (i) pressure, (ii) velocity, and (iii) elevation
3. The power input required by the electric motor that drives this pump (kW)
4. The annual cost required to operate this pump if it runs for 8 hr/day for a total of 260 days and the cost of electrical energy is \$0.12/kWh

Example-2: Pump

A Bell & Gossett Series e-1531 end-suction pump is to be selected for the system shown in Figure below. The system is designed to move water at 60°F from the lower tank to the upper tank. The system must deliver a minimum of 500 gpm of water from the lower tank to the upper tank when the pump is operating. A tilting disc check valve ($\alpha = 5°$) is used to prevent the water from flowing backward when the pump is shut off. The total length of straight pipe in this system is 150 ft. The pipe is 4-nom sch 40 commercial steel.

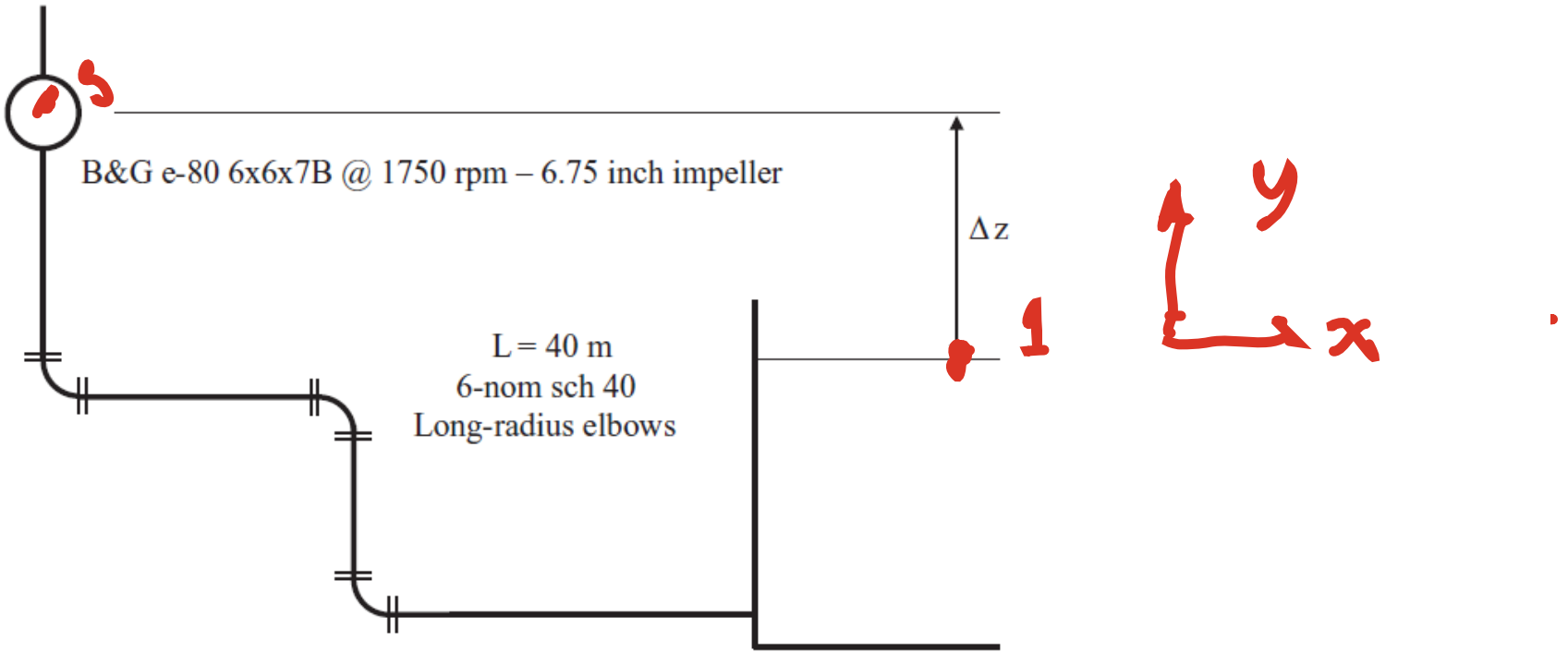
1. Plot the system curve up to 800 gpm.
2. Select a Series e-1531 Bell & Gossett pump for this system.
3. What is the operating point of the pump/pipe system based on the pump you selected in (b)? Specify the capacity (gpm), head (ft), brake horsepower input to the pump (hp), and efficiency of the pump.



Example-3: Pump - Cavitation

A Bell & Gossett Series e-80 6×6×7B pump operating at 1750 rpm with a 6.75-inch impeller is being used to transport water from a large tank as shown in Figure below. The water is at a temperature of 20°C and is being delivered at a rate of 160 m^3/h. The suction line is 6-nom sch 40 commercial steel pipe that is 40 m long. Atmospheric pressure is 1 bar.

1. Determine the maximum height above the water surface that the pump can be installed to prevent cavitation (m).
2. Investigate the effect of the water temperature by plotting the maximum height as a function of the water temperature for $10°C < T < 80°C$



Example-4: Pump - Affine law

A centrifugal pump is capable of delivering 50 L/s of fluid at a head of 18 m while operating at 3550 rpm with a 7-inch impeller. At this condition, the pump power draw is 12 kW and the pump efficiency is 68%. Use the affinity laws to predict the pump capacity, head, power draw, and efficiency if the impeller is changed to a diameter of 9-inches and the operational speed is adjusted to 1150 rpm.

Example-5: Pump - Affine law

A Bell & Gossett Series e-80 4×4×13.5 pump has a 12.5-inch impeller and is operating at 1750 rpm. The pump is installed in a closed-loop pipe system. An analysis of the pump/pipe system at 400 gpm indicates a required pump head of 75 ft. Determine the operating point of the pump/pipe system under these conditions.
After several years of service, engineers have decided to refit the pump with a 12-inch impeller. The engineers also decide to install a variable speed drive for the pump. The new impeller is installed and the operating speed is reduced to 1480 rpm.

1. Use the affinity laws to determine the new pump capacity and head for the operating point determined earlier.
2. Determine the operating point of the pump/pipe system under the new operating conditions (capacity and head).
3. Explain why the points determined in parts (1) and (3) do not match.